# Lecture 12: ODEs and PDEs

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# From derivatives to differential equations

1. **Newton's Laws**:
    1. **First Law (Inertia)**: An object remains in uniform motion unless acted upon by a force.
    2. **Second Law (Force and Acceleration)**: The force acting on an object is equal to the mass of that object times its acceleration ($F=ma$).
    3. **Third Law (Action and Reaction)**: For every action, there is an equal and opposite reaction.

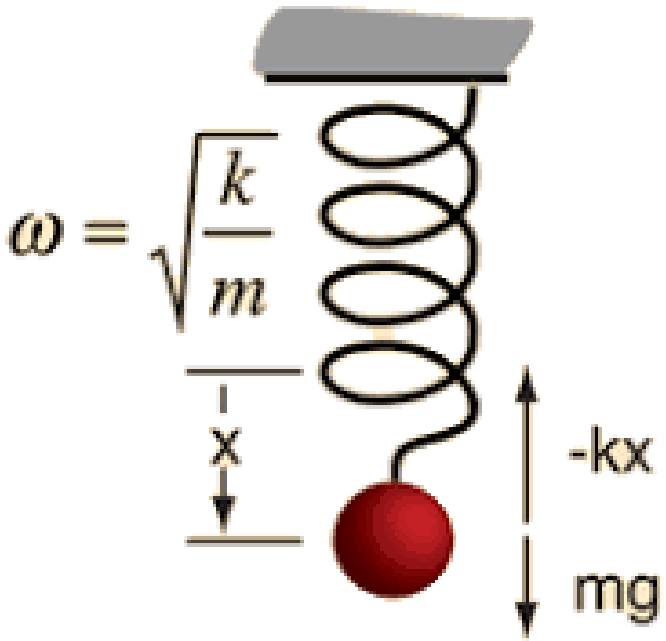
2. Acceleration ($a$) is the second derivative of position ($x$) with respect to time ($t$)

3. **Equation of motion:** $F= m\, dx^2/d^2t$.

4. **Example: Simple Harmonic Motion (SHM)**:
    1. Mass on a spring: $F = - k\, x$
    2. Differential equation for SHM:

    $dt^2/d^2x+\omega^2x=0$,

    where $\omega$ is the angular frequency.




Hooke's Law:

$$F_{spring} = -kx$$

http://hyperphysics.phy-astr.gsu.edu/hbase/shm2.html

# Initial and boundary value problems

Find: $\mathbf{z}(t) = [x(t), y(t), \dot{x}(t), \dot{y}(t)]$

Boundry Conditions:

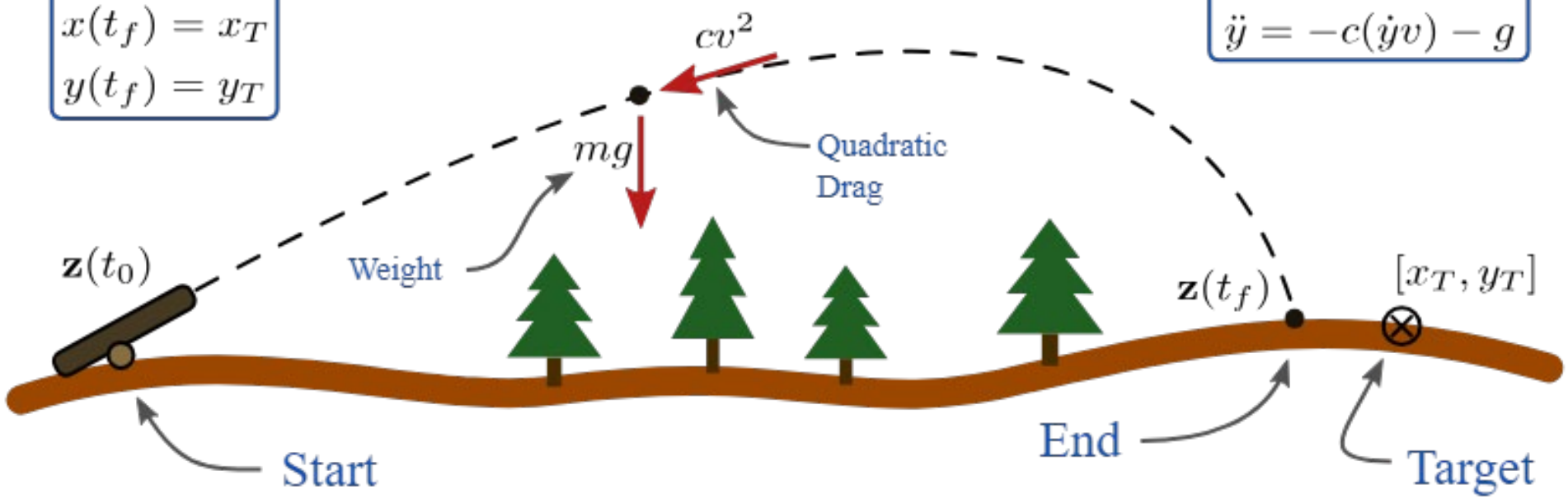
$$x(t_0) = 0$$
$$y(t_0) = 0$$
$$x(t_f) = x_T$$
$$y(t_f) = y_T$$

Cost Function:

$$J = \dot{x}(t_0)^2 + \dot{y}(t_0)^2$$

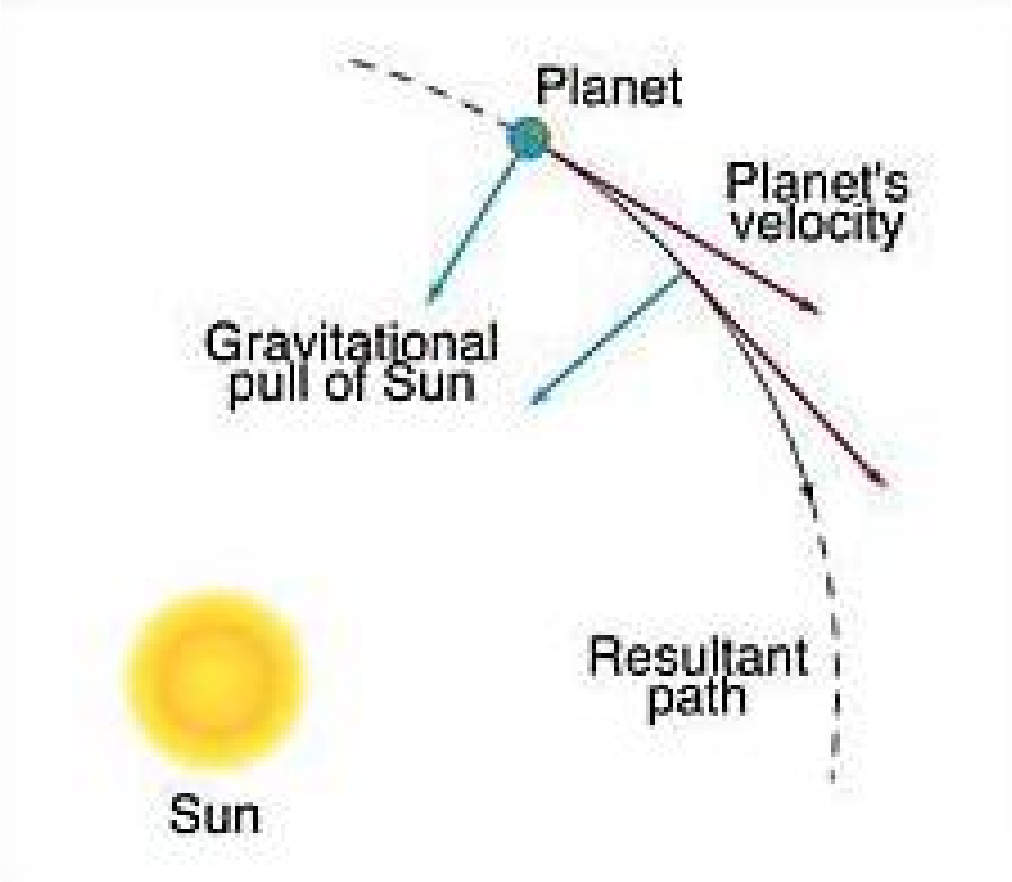
Dynamics:

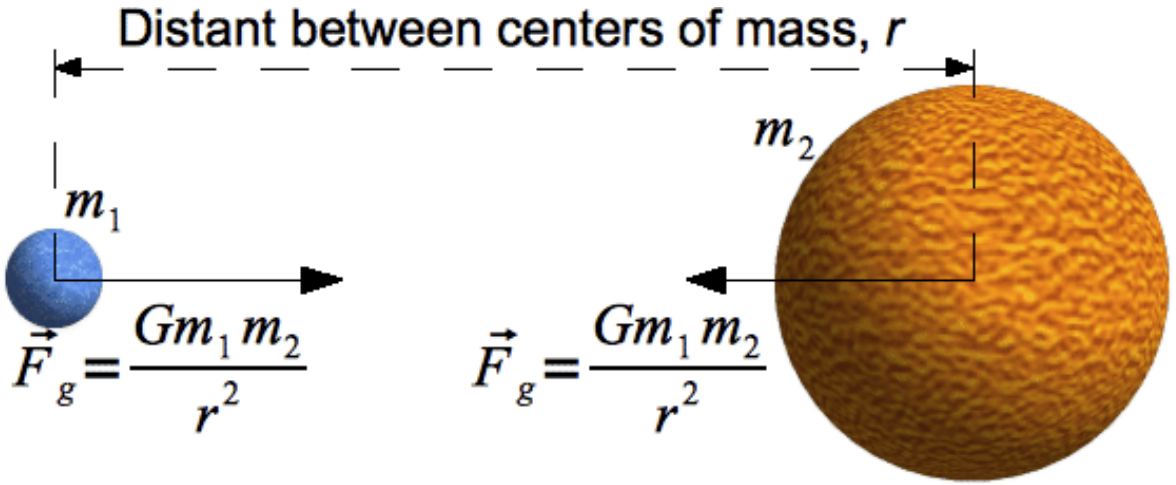
$$v = \sqrt{\dot{x}^2 + \dot{y}^2}$$
$$\ddot{x} = -c(\dot{x}v)$$
$$\ddot{y} = -c(\dot{y}v) - g$$




https://www.matthewpeterkelly.com/tutorials/trajectoryOptimization/canon.html

# Stationary solutions




https://www.phy.olemiss.edu/~luca/astr/Topics-Introduction/Newton-N.html

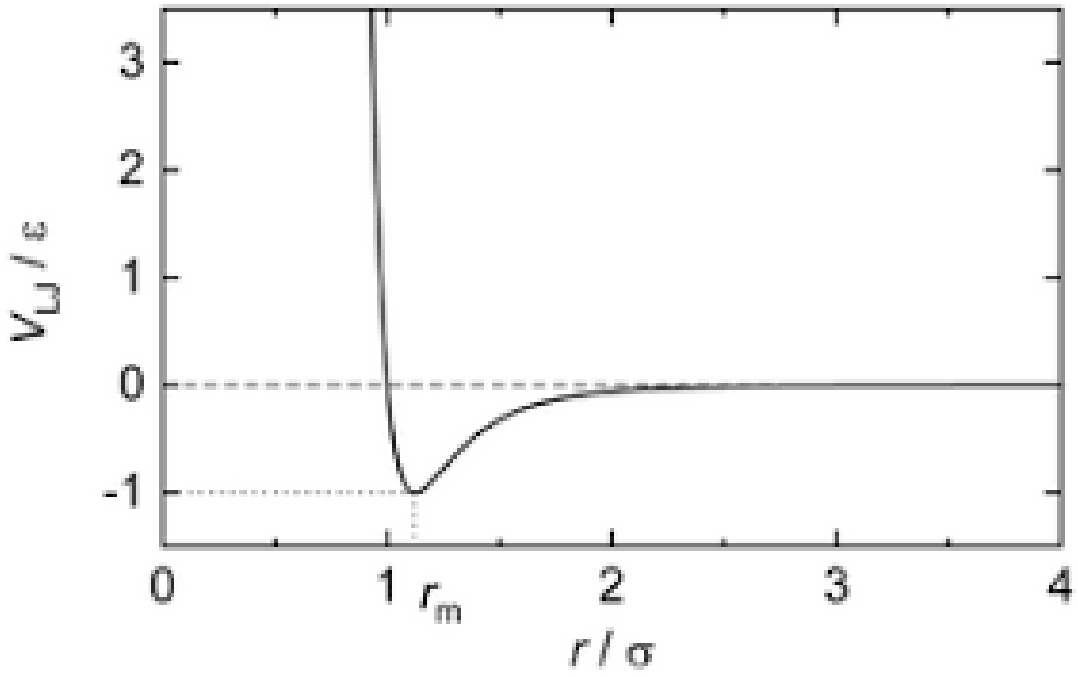
https://erikajanesite.wordpress.com/2017/09/24/225/

# Molecular dynamics

- System of N particles with a pair potential
- Newton's equations of motion (classical *N*-body problem)

$$m\ddot{\mathbf{r}}_\mathbf{i} = -\sum_j \nabla_i V_{\mathrm{LJ}}^{ij}(|\mathbf{r}_\mathbf{i} - \mathbf{r}_\mathbf{j}|)$$

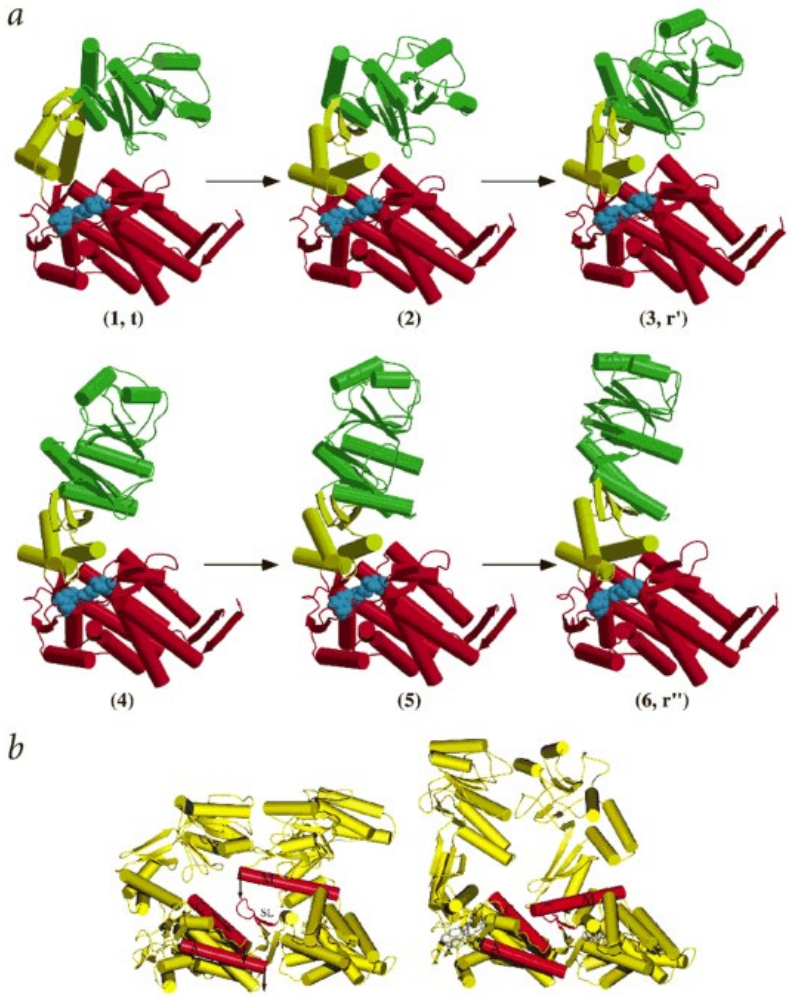
- Box simulation
  - Periodic boundary conditions
  - Minimum-image convention




- If *N* is large enough, system can be characterized by macroscopic parameters
  - Energy-Volume-Number (UVN), microcanonical ensemble
  - Temperature-Volume-Number (TVN), canonical ensemble

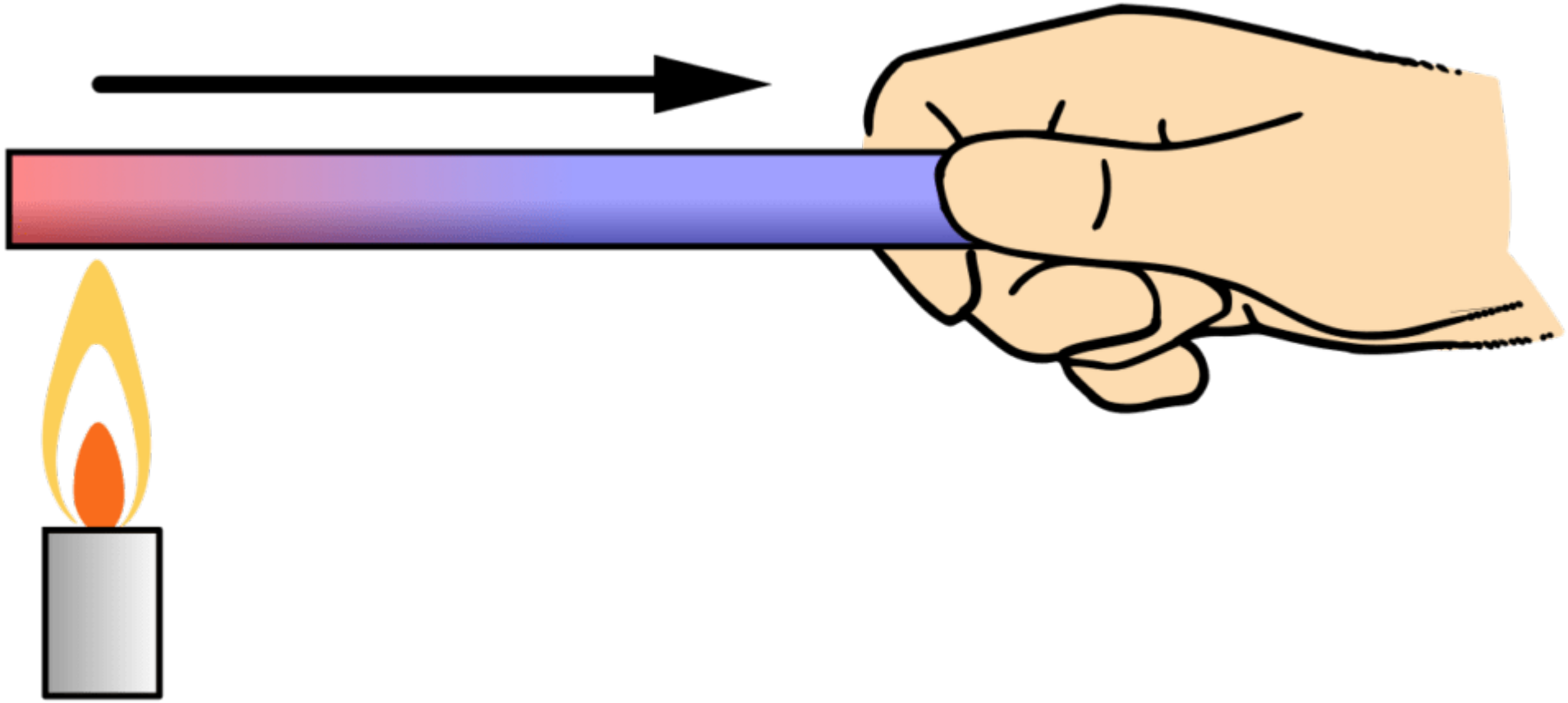
- MD simulations give access to the **equation of state**

# Molecular dynamics




https://www.nature.com/articles/nsb0902-646

# Heat Transfer



https://chemicalengineeringworld.com/modes-of-heat-transfer/

# Heat Transfer

**1. Fourier's Law**: The rate of heat transfer ($q$) through a material is proportional to the negative gradient of the temperature field ($\nabla T$) and the area ($A$) perpendicular to the direction of heat transfer, $q=-kA\nabla T$ where $k$ is the thermal conductivity of the material

**2. Conservation of Energy**: For a given volume, the change in internal energy ($U$) over time ($t$) must equal the net heat flow into the volume minus the work done by the volume on its surroundings. In the absence of work and assuming constant density ($\rho$), this yields:

$$\partial U/\partial t = -\nabla \cdot q + \dot{q}$$

where $\dot{q}$ is the rate of heat generation per unit volume, and $q$ is the heat flux vector

**3. Relating Internal Energy to Temperature**: Assuming the material's specific heat capacity ($c_p$) is constant, the internal energy change can be related to the temperature change:

$$U=\rho c_p T$$

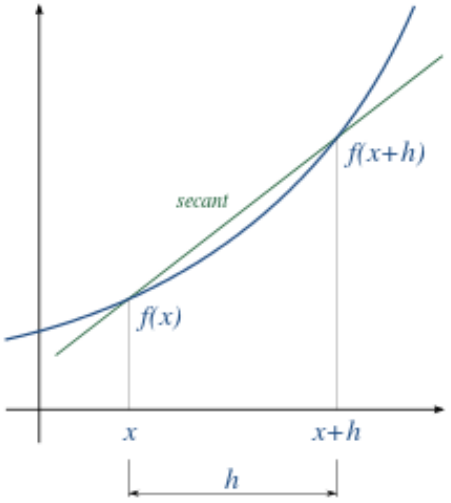
1.Substituting this into the conservation of energy equation and using Fourier's law, we get the heat equation for a homogeneous, isotropic material without internal heat generation as:

$$\rho c_p \partial T/\partial t = k\nabla^2 T$$

# Numerical differentiation

Generic problem: evaluate

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$




We need numerical differentiation when

- Function f is known at a discrete set of points
- Too expensive/cumbersome to do directly
  - E.g. when f(x) itself is a solution to a complex web of non-linear equations, calculating f'(x) explicitly will require rewriting all the equations

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# Forward difference

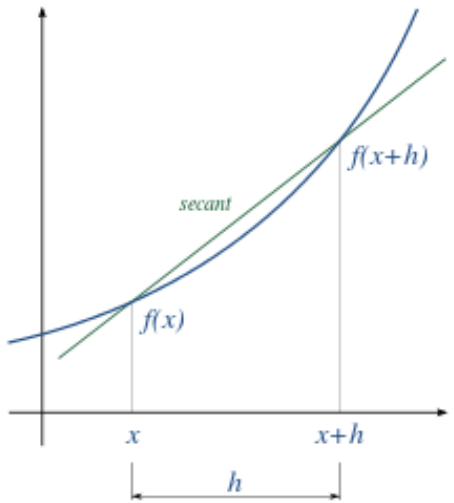
Simply approximate

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

by

$$\frac{\mathrm{d}f}{\mathrm{d}x} \simeq \frac{f(x+h) - f(x)}{h}$$

where *h* is finite




Taylor theorem:

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2} f''(x) + \ldots$$

gives the approximation error estimate of

$$R_{\text{forw}} = -\frac{1}{2} h f''(x) + \mathcal{O}(h^2)$$

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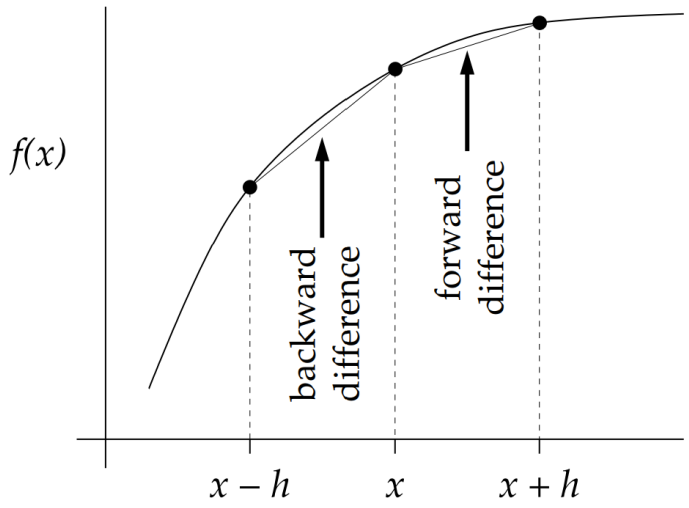
# Backward difference

Backward difference

$$\frac{\mathrm{d}f}{\mathrm{d}x} \simeq \frac{f(x) - f(x-h)}{h}$$




Taylor theorem:

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) + \ldots$$

gives the approximation error estimate of

$$R_{\text{back}} = \frac{1}{2}hf''(x) + \mathcal{O}(h^2)$$

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# Central difference

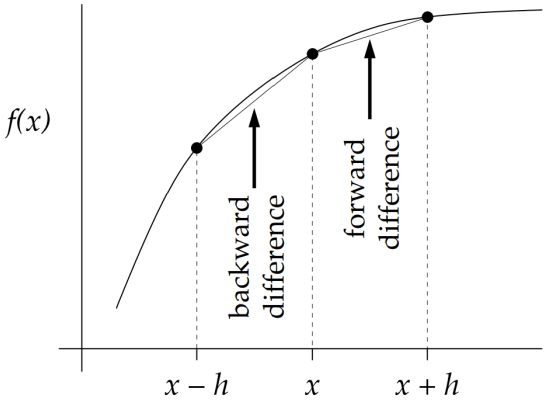
Recall the forward and backward difference and their errors

$$\frac{\mathrm{d}f}{\mathrm{d}x} \simeq \frac{f(x+h)-f(x)}{h}$$

$$R_{\text{forw}} = -\frac{1}{2}hf''(x) + \mathcal{O}(h^2)$$

$$\frac{\mathrm{d}f}{\mathrm{d}x} \simeq \frac{f(x)-f(x-h)}{h}$$

$$R_{\text{back}} = \frac{1}{2}hf''(x) + \mathcal{O}(h^2)$$




Taking the average of the two cancels out the *O(h)* error term

**central difference**

$$\frac{df}{dx} \simeq \frac{f(x+h)-f(x-h)}{2h}$$

Error estimate:

$$R_{\text{cent}} = -\frac{f'''(x)}{6}h^2 + \mathcal{O}(h^3)$$

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# High-order central difference

To improve the approximation error use more than two function evaluations, e.g.

$$\frac{df}{dx} \simeq \frac{Af(x+2h) + Bf(x+h) + Cf(x) + Df(x-h) + Ef(x-2h)}{h} + O(h^4)$$

Determine *A,B,C,D,E* using Taylor expansion to cancel all terms up to $h^4$

$$\frac{df}{dx} \simeq \frac{-f(x+2h) + 8f(x+h) - 8f(x-h) + f(x-2h)}{12h} + \frac{h^4}{30} f^{(5)}(x)$$

High-order terms:

| Derivative | Accuracy | −5 | −4 | −3 | −2 | −1 | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | | | | | −1/2 | 0 | 1/2 | | | | |
| | 4 | | | | 1/12 | −2/3 | 0 | 2/3 | −1/12 | | | |
| | 6 | | | −1/60 | 3/20 | −3/4 | 0 | 3/4 | −3/20 | 1/60 | | |
| | 8 | | 1/280 | −4/105 | 1/5 | −4/5 | 0 | 4/5 | −1/5 | 4/105 | −1/280 | |

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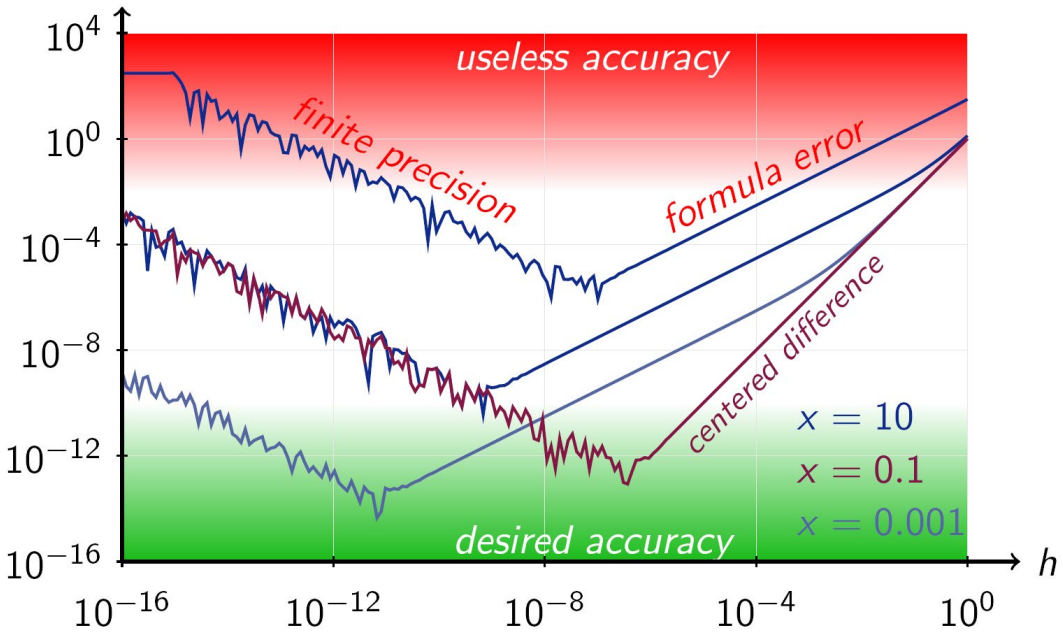
# Balancing truncation and round-off errors

If *h* is too small, round-off errors become important

- cannot distinguish *x* and *x+h* and/or *f(x+h)* and *f(x)* with enough accuracy




Credit: Wikipedia

As a rule of thumb, if $\varepsilon$ is machine precision and the truncation error is of order $O(h^n)$, then *h* should not be much smaller than $h \sim \sqrt[n+1]{\varepsilon}$

The higher the finite difference order is, the larger *h* should be

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# Balancing truncation and round-off errors

Let f(x) = exp(x)

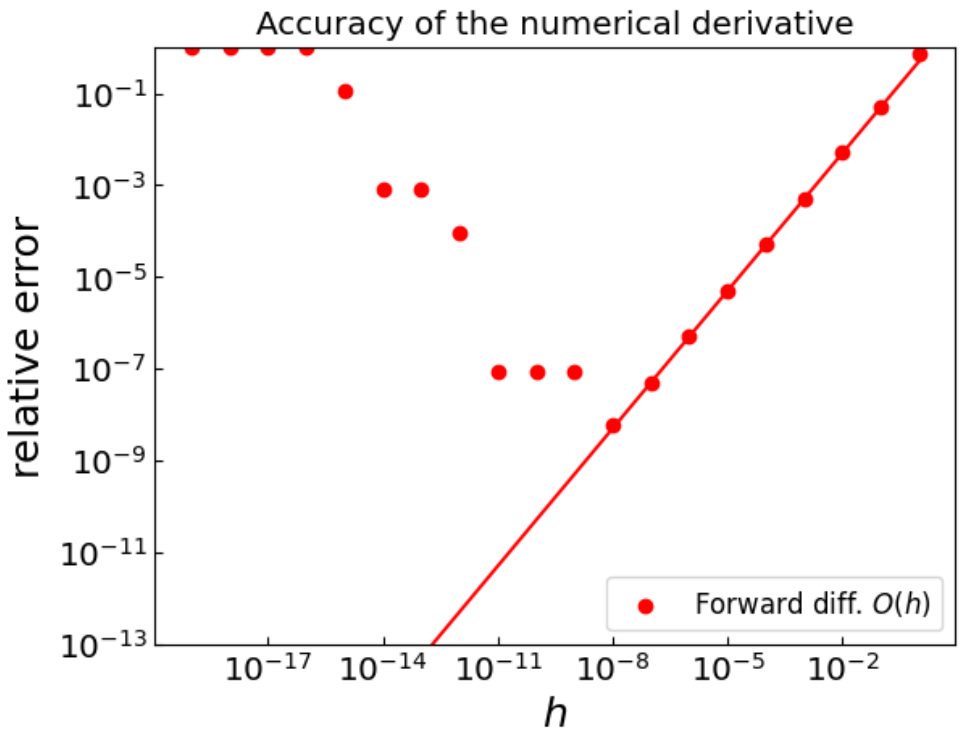
Calculate the derivatives at x = 0

```
def f(x):
    return np.exp(x)

def df(x):
    return np.exp(x)
```

**Forward difference O(h):**

Optimal $h \sim \sqrt[2]{10^{-16}} \sim 10^{-8}$

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# Balancing truncation and round-off errors

Let f(x) = exp(x)

Calculate the derivatives at x = 0

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def f(x):
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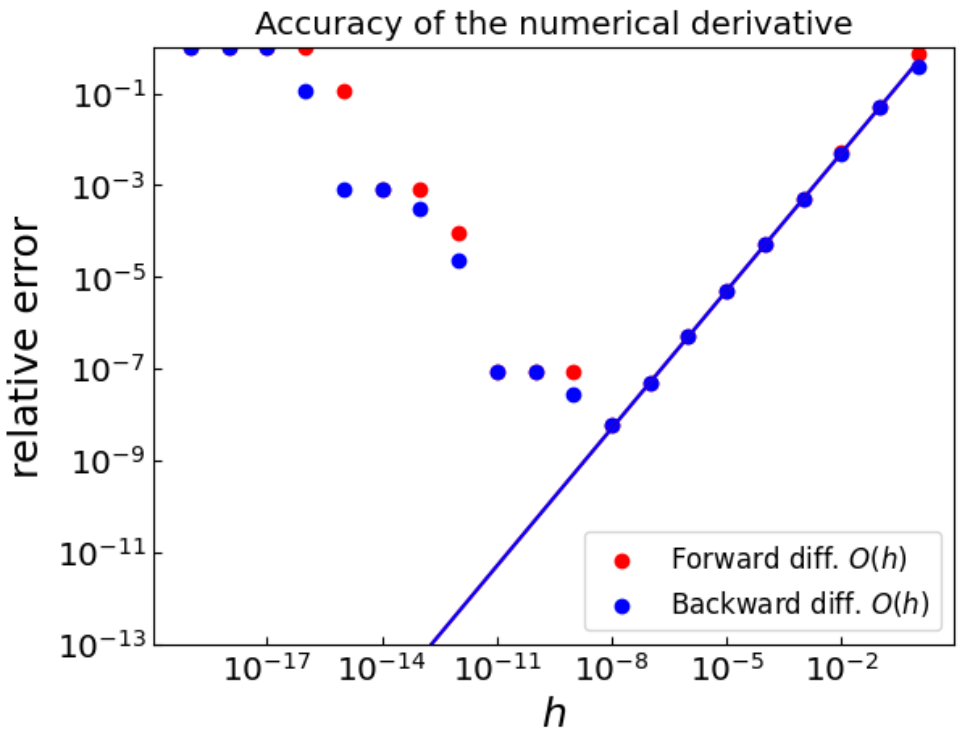
```

**Backward difference O(h):**

Optimal $h \sim \sqrt[2]{10^{-16}} \sim 10^{-8}$

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# Balancing truncation and round-off errors

Let f(x) = exp(x)

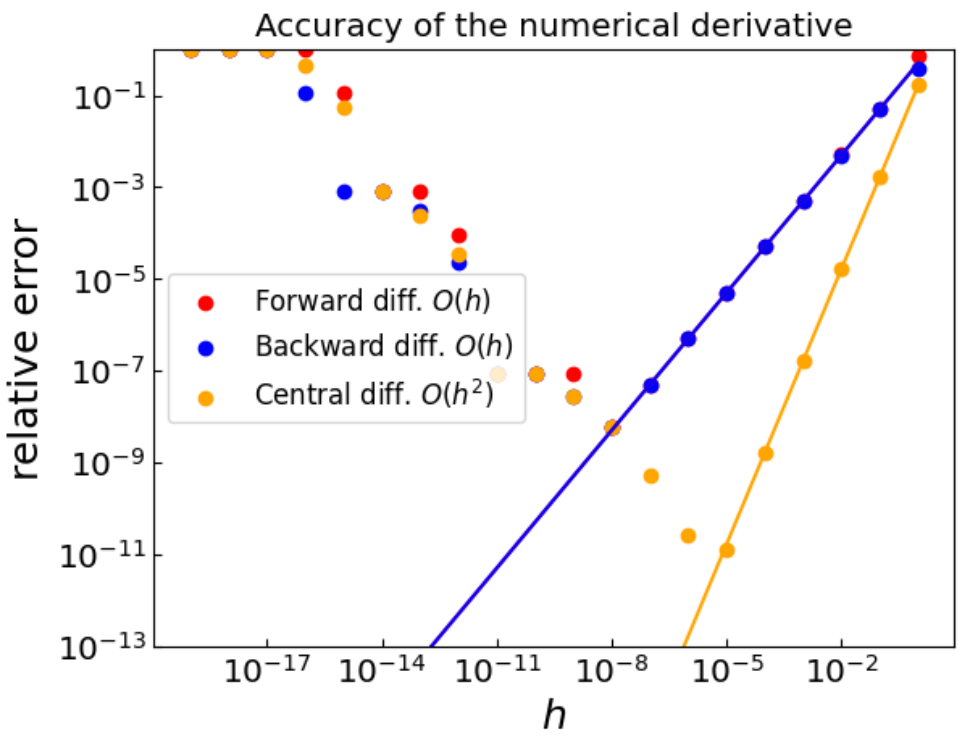
Calculate the derivatives at x = 0

```
def f(x):
    return np.exp(x)

def df(x):
    return np.exp(x)
```

**Central difference O(h²):**

Optimal $h \sim \sqrt[3]{10^{-16}} \sim 10^{-5}$

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# Balancing truncation and round-off errors

Let f(x) = exp(x)

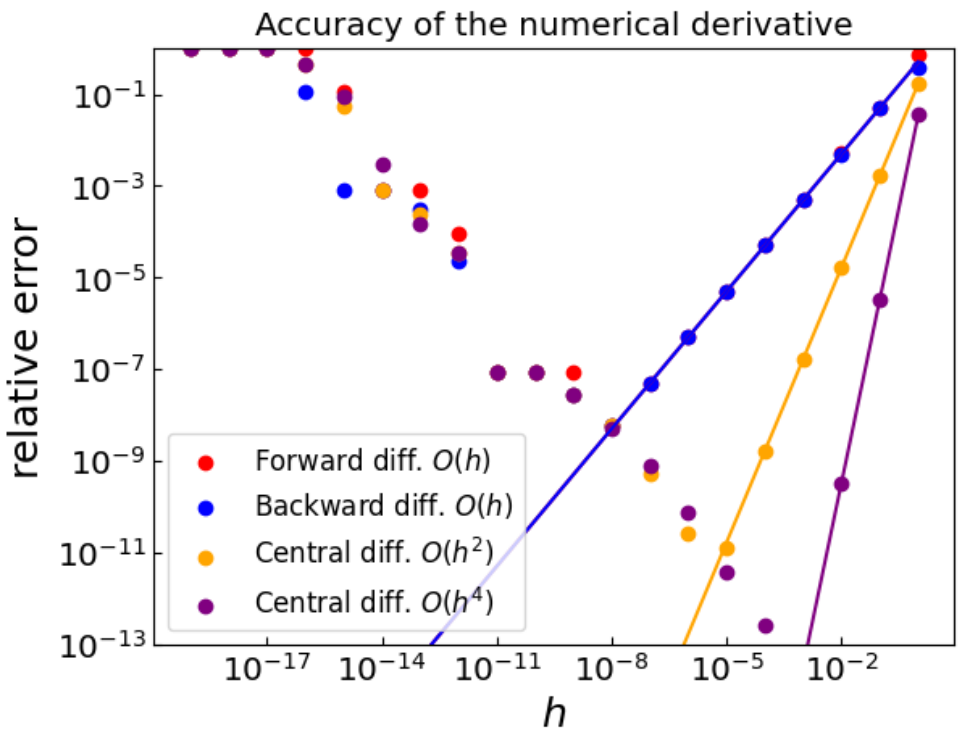
Calculate the derivatives at x = 0

```
def f(x):
    return np.exp(x)

def df(x):
    return np.exp(x)
```

**Central difference O(h⁴):**

Optimal $h \sim \sqrt[5]{10^{-16}} \sim 10^{-3}$

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# High-order derivatives

Central difference

$$\frac{df}{dx}(x) \simeq \frac{f(x+h/2) - f(x-h/2)}{h}$$

Now apply the central difference again to *f'(x+h/2)* and *f'(x-h/2)*

$$\begin{aligned} f''(x) &\simeq \frac{f'(x+h/2) - f'(x-h/2)}{h} \\ &= \frac{[f(x+h) - f(x)]/h - [f(x) - f(x-h)]/h}{h} \\ &= \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}. \end{aligned}$$

General formula (to order *h*)

$$f^{(n)}(x) = \frac{1}{h^n} \sum_{k=0}^{n} (-1)^k \binom{n}{k} f\left[x + (n/2 - k)h\right] + O(h^2)$$

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# Second derivative

```
def d2f_central(f,x,h):
    return (f(x+h) - 2*f(x) + f(x-h)) / (h**2)
```

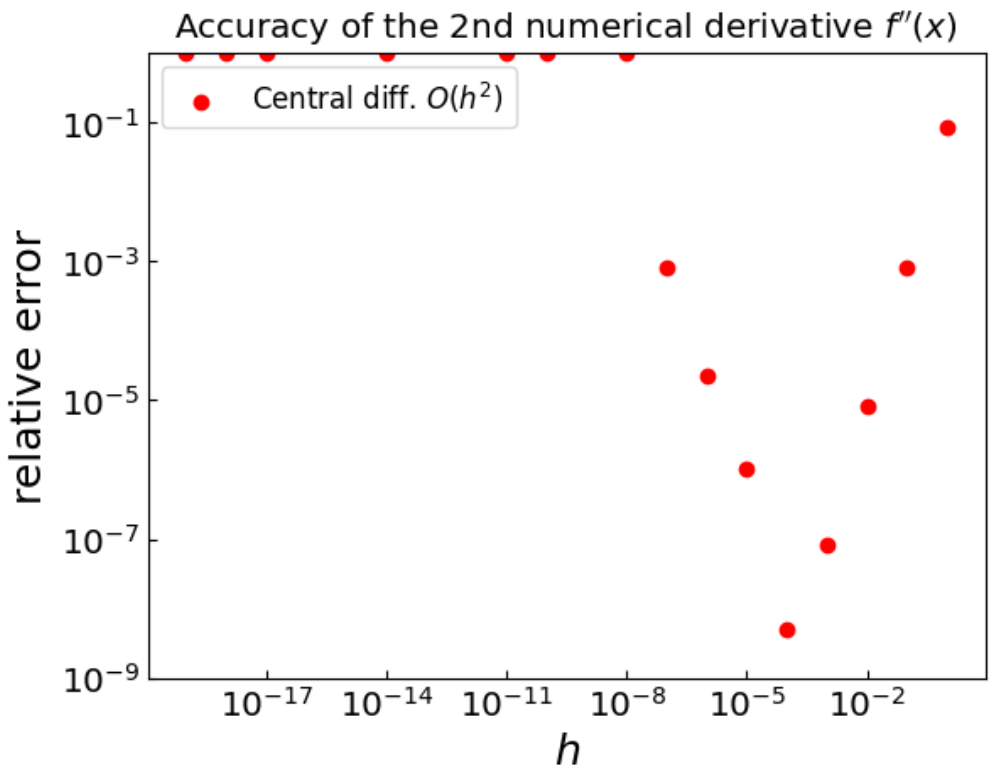
f(x) = exp(x)

```
def f(x):
    return np.exp(x)

def df(x):
    return np.exp(x)

def d2f(x):
    return np.exp(x)
```

Optimal $h \sim \sqrt[4]{10^{-16}} \sim 10^{-4}$

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# Partial derivatives

Let us have *f(x,y)*
Use central difference to calculate first-order derivatives

$$\frac{\partial f}{\partial x} = \frac{f(x+h/2,y) - f(x-h/2,y)}{h}$$
$$\frac{\partial f}{\partial y} = \frac{f(x,y+h/2) - f(x,y-h/2)}{h}$$

Reapply the central difference to calculate $\partial^2 f(x,y)/\partial x \partial y$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{f(x+h/2,y+h/2) - f(x-h/2,y+h/2) - f(x+h/2,y-h/2) + f(x-h/2,y-h/2)}{h^2}$$

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# Summary: Numerical differentiation

- Forward/backward differences
  - Useful when we are given a grid of function values
  - Have limited accuracy (linear in *h*)

- Central difference
  - More precise than forward/backward differences (quadratic in *h*)
  - Gives f'(x) estimate at the midpoint of function evaluation points

- Higher-order formulas are obtained by using more than two function evaluations
  - Can be used when limited number of function evaluations available

- Straightforwardly extendable to high-order and partial derivatives

- Balance between truncation and round-off error must be respected
  - *h* should not be taken too small

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# Numerical derivative and ordinary differential equations

Ordinary differential equation

$$\frac{dx}{dt} = f(x, t),$$

with initial condition

$$f(x, t_0) = f_0$$

Use the forward difference to approximate dx/dt

$$\frac{dx}{dt} \approx \frac{x(t+h) - x(t)}{h}$$

gives the **Euler method** of solving the equation for x(t)

$$x(t+h) = x(t) + h\, f[x(t), t]$$

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# Ordinary Differential Equations (ODE)

First-order ordinary differential equation (ODE) is an equation of the form

$$\frac{dx}{dt} = f(x, t),$$

with initial condition

$$x(t = 0) = x_0$$

This determines the x(t) dependence at t>0.

In many physical applications *t* plays the role of the time variable (classical mechanics problems), although this is not always the case.

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# When we need numerical methods for ODEs

The solution to an ODE

$$\frac{dx}{dt} = f(x, t), \qquad x(t = 0) = x_0$$

can formally be written as

$$x(t) = x_0 + \int_0^t f[x(t'), t'])dt'$$

If $f$ does not depend on $x$, the solution can be obtained through (numerical) integration

In some other cases the solution can be obtained through the separation of variables, e.g.

$$\frac{dx}{dt} = \frac{2x}{t}$$

In all other cases, the solution has to be obtained numerically.

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# Numerical methods for ODEs

Typically obtain the solution by taking small steps from *x(t)* to *x(t+h)*

**Characteristics:**

- Explicit or implicit
  - Explicit methods: use *x(t)* to calculate *x(t+h)* directly
  - Implicit methods: have to solve a (non-linear) equation for *x(t+h)*

- Accuracy
  - Truncation error at each step is of order $O(h^n)$
  - Some schemes are explicitly time-reversal and/or conserve energy
  - Adaptive methods adjust the step size *h* to control the error to the desired accuracy

- Stability
  - Whether the accumulated error is bounded (that's where implicit methods shine)

- Consistency
  - Consistent methods reproduce the exact solution in the limit $h \to 0$

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# Euler's method

$$\frac{dx}{dt} = f(x, t),$$

Let us apply the Taylor expansion to express *x(t+h)* in terms of *x(t)*:

$$x(t+h) = x(t) + h\frac{dx}{dt} + O(h^2)\ .$$

Given that dx/dt = f(x,t) and neglecting the high-order terms in *h* we have

$$x(t+h) \approx x(t) + h\,f[x(t), t] \qquad \textit{Euler method}$$

We can iteratively apply this relation starting from t = 0 to evaluate x(t) at t > 0.

This is the essence of the **Euler method** -- the simplest method for solving ODEs numerically.

**Error:**

- Local (per time step): $O(h^2)$
- Global ($N=t_{end}/h$ time steps): $O(h)$

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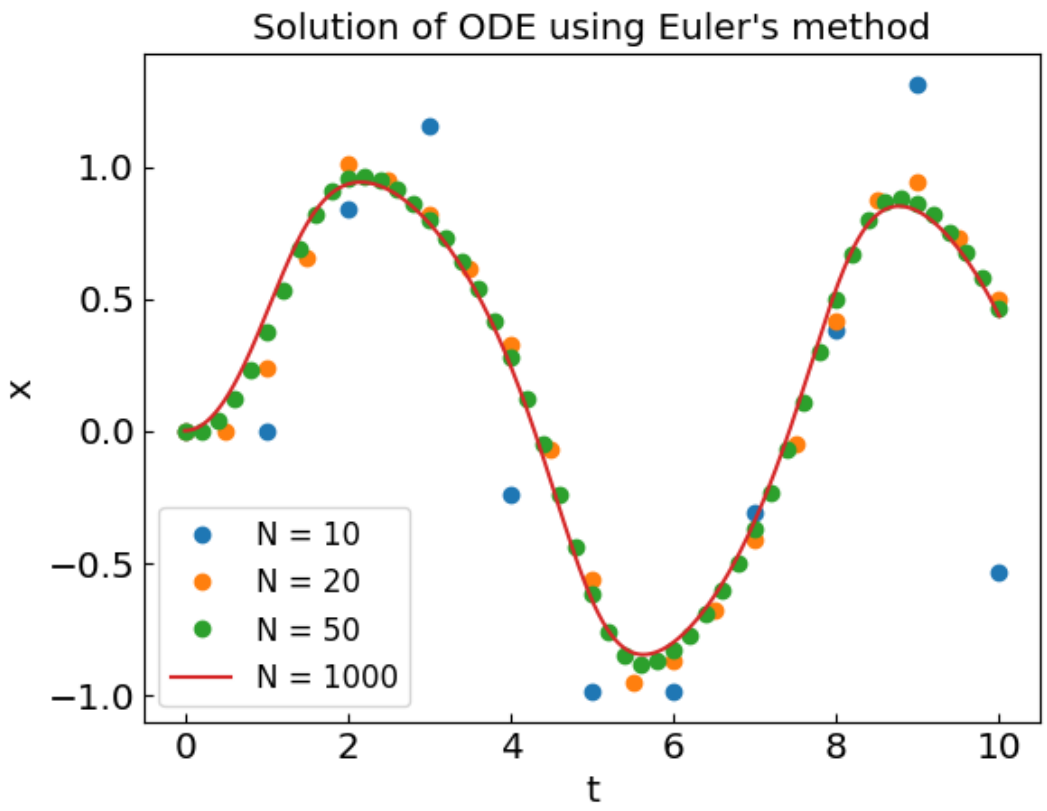
# Euler's method

```python
import numpy as np

def ode_euler_step(f, x, t, h):
    """Perform a single step h using Euler's scheme.

    Args:
    f: the function that defines the ODE.
    x: the value of the dependent variable at the present step.
    t: the present value of the time variable.
    h: the time step

    Returns:
    xnew: the value of the dependent variable at the step t+h
    """
    return x + h * f(x,t)

def ode_euler(f, x0, t0, h, nsteps):
    """Solve an ODE dx/dt = f(x,t) from t = t0 to t = t0 + h*steps using Euler's method.

    Args:
          f: the function that defines the ODE.
         x0: the initial value of the dependent variable.
         t0: the initial value of the time variable.
          h: the time step
     nsteps: the total number of Euler steps

    Returns:
    t,x: the pair of arrays corresponding to the time and dependent variables
    """

    t = np.zeros(nsteps + 1)
    x = np.zeros(nsteps + 1)
    x[0] = x0
    t[0] = t0
    for i in range(0, nsteps):
        t[i + 1] = t[i] + h
        x[i + 1] = ode_euler_step(f, x[i], t[i], h)
    return t,x
```

$$\frac{dx}{dt} = -x^3 + \sin t, \qquad x(t=0) = 0.$$

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# Midpoint method (2nd order Runge-Kutta)

Euler's method essentially corresponds to approximating the derivative dx/dt with a *forward difference*

$$\frac{dx}{dt} = f(x,t) \approx \frac{x(t+h) - x(t)}{h} + \mathcal{O}(h).$$

Recall that central (midpoint) difference gives better accuracy

$$f(x, t+h/2) \approx \frac{x(t+h) - x(t)}{h} + \mathcal{O}(h^2).$$

therefore

$$x(t+h) = x(t) + hf[x(t+h/2), t+h/2)] + \mathcal{O}(h^3)$$

How to calculate *x(t+h/2)* in r.h.s? Use Euler's method $x(t+h/2) = x(t) + \frac{1}{2}hf(x,t) + \mathcal{O}(h^2)$

Therefore, $x(t+h) = x(t) + hf\left[x(t) + \frac{1}{2}hf(x,t), t + \frac{1}{2}h\right] + \mathcal{O}(h^3)$ , which can be written in two steps

$$\begin{aligned} k_1 &= h\,f(x,t), && \textit{trial step} \\ k_2 &= h\,f(x + k_1/2, t + h/2), && \textit{real step} \\ x(t+h) &= x(t) + k_2\ . \end{aligned}$$

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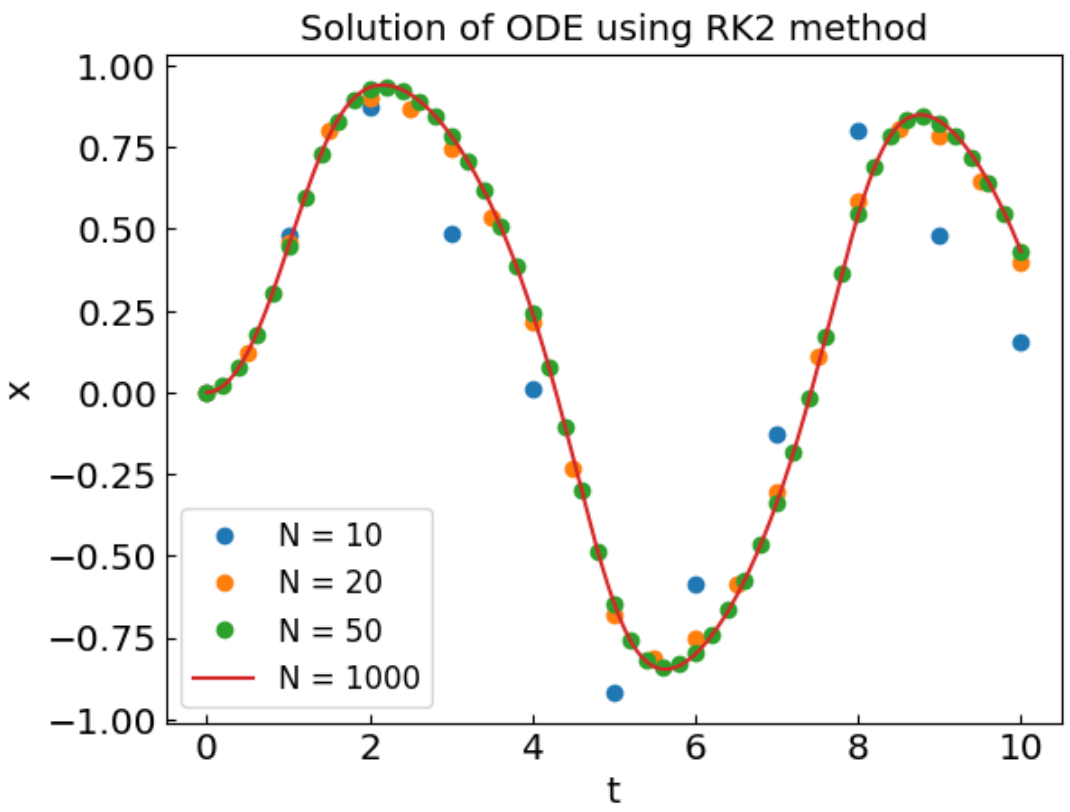
# Midpoint method (2nd order Runge-Kutta)

```python
def ode_rk2_step(f, x, t, h):
    """Perform a single step h using 2nd order Runge-Kutta scheme.

    Args:
    f: the function that defines the ODE.
    x: the value of the dependent variable at the present step.
    t: the present value of the time variable.
    h: the time step

    Returns:
    xnew: the value of the dependent variable at the step t+h
    """
    k1 = h * f(x,t)
    k2 = h * f(x + k1/2., t + h /2.)
    return x + k2

def ode_rk2(f, x0, t0, h, nsteps):
    """Solve an ODE dx/dt = f(x,t) from t = t0 to t = t0 + h*steps using Euler's method.

    Args:
          f: the function that defines the ODE.
         x0: the initial value of the dependent variable.
         t0: the initial value of the time variable.
          h: the time step
     nsteps: the total number of Euler steps

    Returns:
    t,x: the pair of arrays corresponding to the time and dependent variables
    """

    t = np.zeros(nsteps + 1)
    x = np.zeros(nsteps + 1)
    x[0] = x0
    t[0] = t0
    for i in range(0, nsteps):
        t[i + 1] = t[i] + h
        x[i + 1] = ode_rk2_step(f, x[i], t[i], h)
    return t,x
```

$$\frac{dx}{dt} = -x^3 + \sin t, \qquad x(t=0) = 0.$$




**Error:**

- Local (per time step): $O(h^3)$
- Global ($N=t_{end}/h$ time steps): $O(h^2)$

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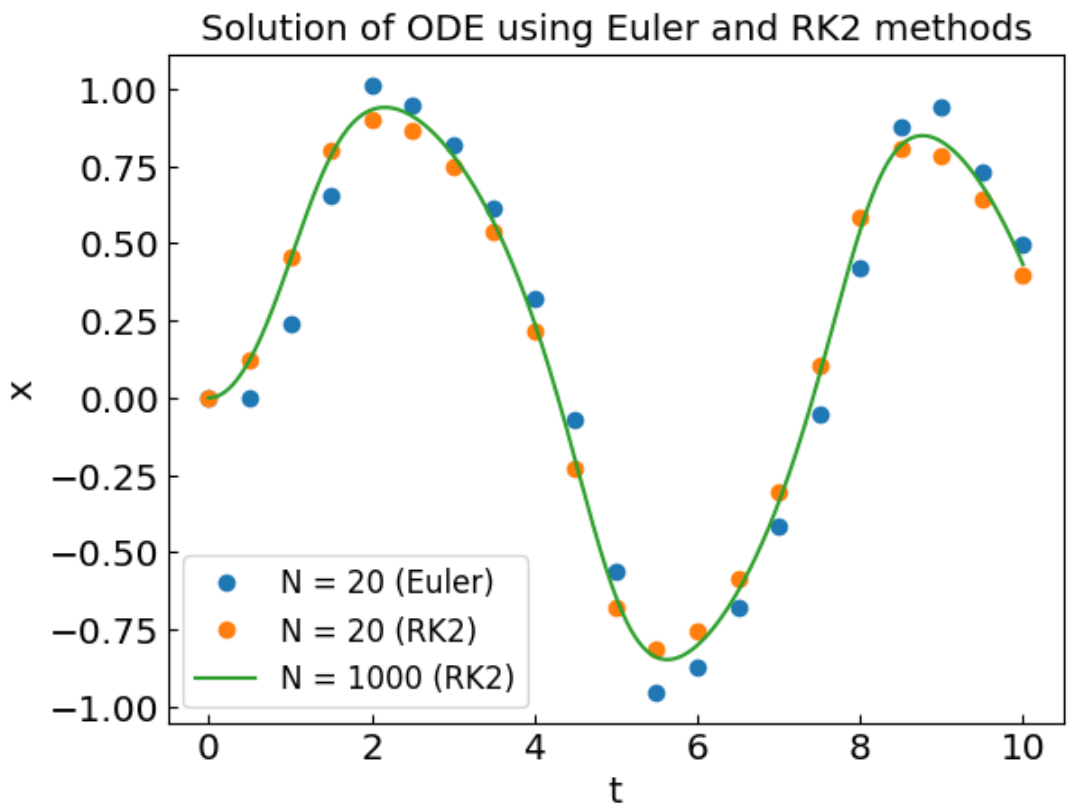
# Midpoint method (2nd order Runge-Kutta)

```python
def ode_rk2_step(f, x, t, h):
    """Perform a single step h using 2nd order Runge-Kutta scheme.

    Args:
    f: the function that defines the ODE.
    x: the value of the dependent variable at the present step.
    t: the present value of the time variable.
    h: the time step

    Returns:
    xnew: the value of the dependent variable at the step t+h
    """
    k1 = h * f(x,t)
    k2 = h * f(x + k1/2., t + h /2.)
    return x + k2

def ode_rk2(f, x0, t0, h, nsteps):
    """Solve an ODE dx/dt = f(x,t) from t = t0 to t = t0 + h*steps using Euler's method.

    Args:
          f: the function that defines the ODE.
         x0: the initial value of the dependent variable.
         t0: the initial value of the time variable.
          h: the time step
     nsteps: the total number of Euler steps

    Returns:
    t,x: the pair of arrays corresponding to the time and dependent variables
    """

    t = np.zeros(nsteps + 1)
    x = np.zeros(nsteps + 1)
    x[0] = x0
    t[0] = t0
    for i in range(0, nsteps):
        t[i + 1] = t[i] + h
        x[i + 1] = ode_rk2_step(f, x[i], t[i], h)
    return t,x
```

$$\frac{dx}{dt} = -x^3 + \sin t, \qquad x(t=0) = 0.$$




**Error:**

- Local (per time step): $O(h^3)$
- Global ($N=t_{end}/h$ time step): $O(h^2)$

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# Classical 4th order Runge-Kutta method

The above logic can be generalized to cancel high-order error terms in various powers in *h,* requiring more and more evaluations of function *f*(x,t) at intermediate steps.

The following classical 4th-order Runge-Kutta method is often considered a sweet spot.

It corresponds to the following scheme:

$$
\begin{aligned}
k_1 &= h\,f(x,t),\\
k_2 &= h\,f(x+k_1/2,\,t+h/2),\\
k_3 &= h\,f(x+k_2/2,\,t+h/2),\\
k_4 &= h\,f(x+k_3,\,t+h),\\
x(t+h) &= x(t) + \frac{1}{6}(k_1+2k_2+2k_3+k_4)\ .
\end{aligned}
$$

**Error:**

- Local (per time step): $O(h^5)$
- Global ($N=t_{end}/h$ time steps): $O(h^4)$

The classical 4th-order Runge-Kutta method is a good first choice for solving physics ODEs.

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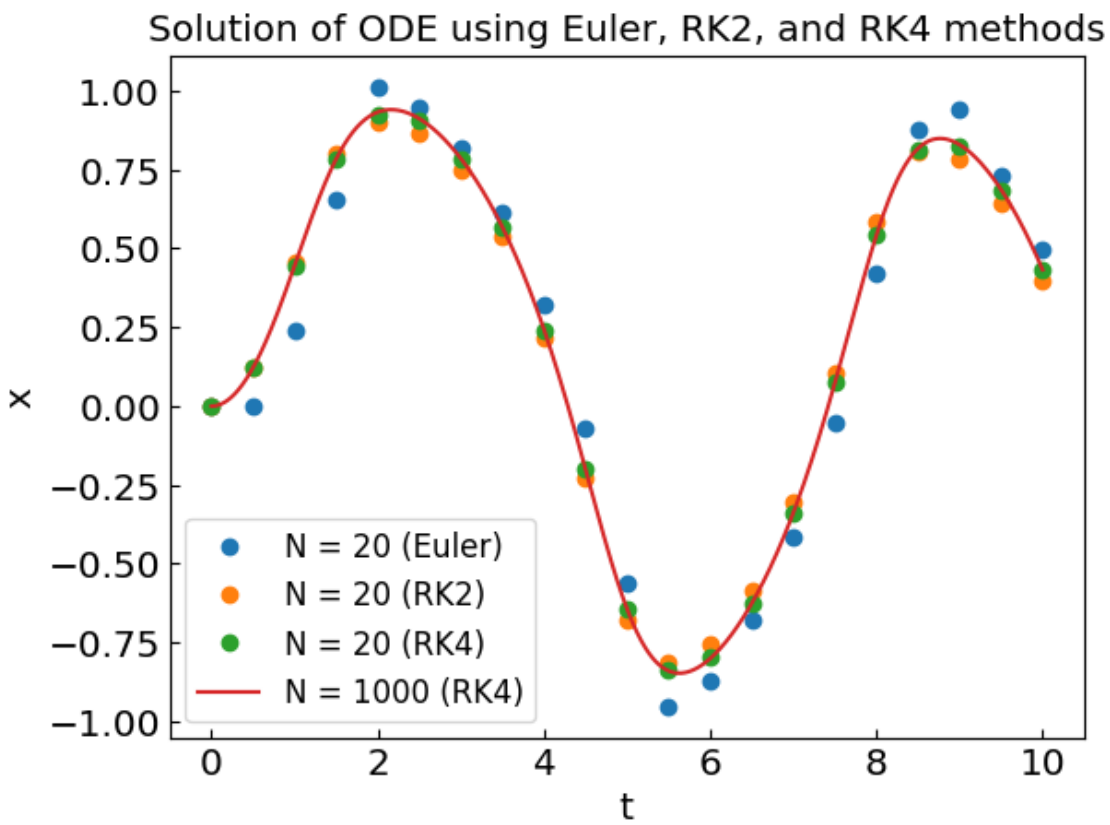
# Classical 4th order Runge-Kutta method

```python
def ode_rk4_step(f, x, t, h):
    """Perform a single step h using 4th order Runge-Kutta method.

    Args:
    f: the function that defines the ODE.
    x: the value of the dependent variable at the present step.
    t: the present value of the time variable.
    h: the time step

    Returns:
    xnew: the value of the dependent variable at the step t+h
    """
    k1 = h * f(x,t)
    k2 = h * f(x + k1/2., t + h /2.)
    k3 = h * f(x + k2/2., t + h /2.)
    k4 = h * f(x + k3, t + h)
    return x + (k1 + 2. * k2 + 2. * k3 + k4) / 6.

def ode_rk4(f, x0, t0, h, nsteps):
    """Solve an ODE dx/dt = f(x,t) from t = t0 to t = t0 + h*steps using 4th order Runge-Kutta method.

    Args:
          f: the function that defines the ODE.
         x0: the initial value of the dependent variable.
         t0: the initial value of the time variable.
          h: the time step
     nsteps: the total number of Euler steps

    Returns:
    t,x: the pair of arrays corresponding to the time and dependent variables
    """

    t = np.zeros(nsteps + 1)
    x = np.zeros(nsteps + 1)
    x[0] = x0
    t[0] = t0
    for i in range(0, nsteps):
        t[i + 1] = t[i] + h
        x[i + 1] = ode_rk4_step(f, x[i], t[i], h)
    return t,x
```

$$\frac{dx}{dt} = -x^3 + \sin t, \qquad x(t=0) = 0.$$

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# Adaptive time step

$$\frac{dx}{dt} = f(x, t)$$

The choice of the time step is important to reach the desired accuracy/performance.

- *h* too large: the desired accuracy not reached
- *h* too small: we waste computing resources on unnecessary iterations
- Local truncation error itself is a function of time depending on the behavior of f(x,t)

**Adaptive time step:** make a local error estimate and adjust *h* to correspond to the desired accuracy

Ways to estimate the error:

- Make two small steps (*h*) and compare x(t+2h) to the one from a single double step *2h*
- Use two methods of a different order and compare their results (e.g. Runge-Kutta-Fehlberg method RKF45)

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# Adaptive time step in RK4 using double step

Recall that the error for one RK4 time step *h* is of order $ch^5$.
Let us take two RK4 steps h to approximate $x(t+2h) \approx x_1$. Then,

$$x(t+2h) \approx x_1 + 2ch^5$$

Now take single RK4 step $x(t+2h) \approx x_2$ of length *2h*

$$x(t+2h) \approx x_2 + 32ch^5$$

The local error estimate for a single RK4 time step h is then

$$\epsilon_{\text{RK4}} = |ch^5| = \frac{|x_1 - x_2|}{30}.$$

If the desired accuracy per unit time is $\delta$, the desired accuracy per time step h' is

$$h'\delta = ch'^5$$

so the time step should be adjusted from h to h' as

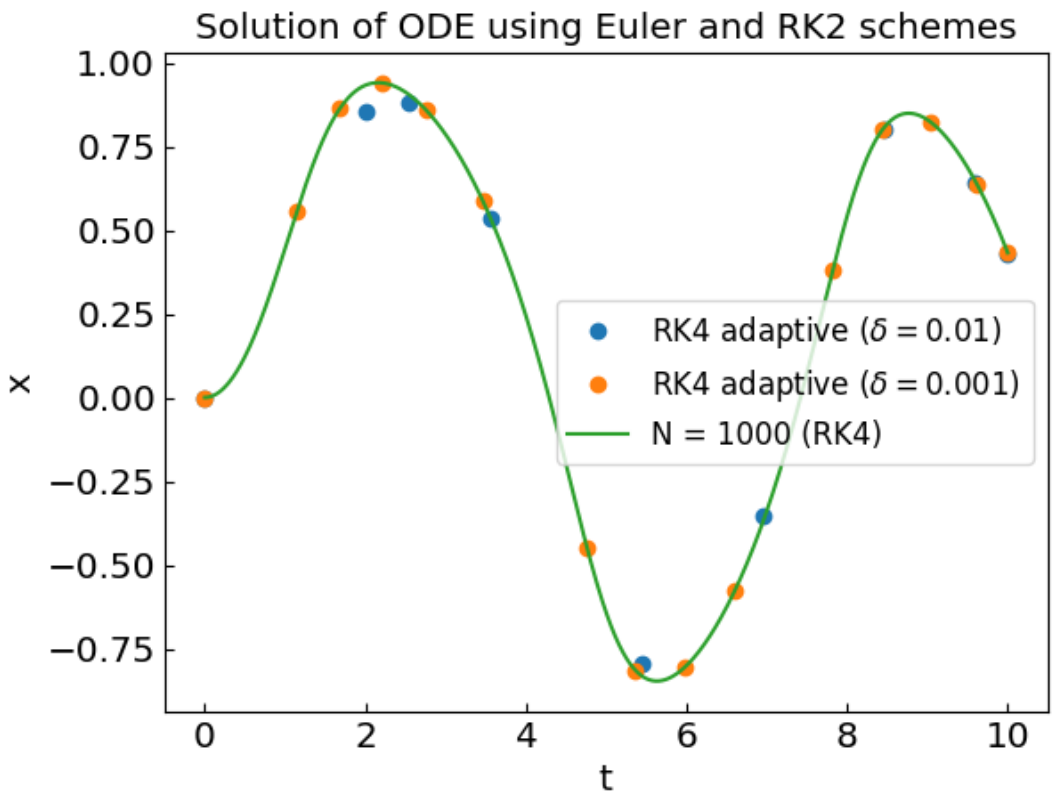
$$h' = h\left(\frac{30h\delta}{|x_1 - x_2|}\right)^{1/4}.$$

- *h'>h*: our step size is too small, move on to x(t+2h) and increase the step size to h'
- *h'<h*: our step size is too large, decrease step size to h' and try the current step again

# RK4 method with adaptive step size

```python
def ode_rk4_adaptive(f, x0, t0, h0, tmax, delta = 1.e-6):

    ts = [t0]
    xs = [x0]
    h = h0
    t = t0
    i = 0
    while (t < tmax):
        if (t + h >= tmax):
            ts.append(tmax)
            h = tmax - t
            xs.append(ode_rk4_step(f, xs[i], ts[i], h))
            t = tmax
            break

        x1 = ode_rk4_step(f, xs[i], ts[i], h)
        x1 = ode_rk4_step(f, x1, ts[i] + h, h)
        x2 = ode_rk4_step(f, xs[i], ts[i], 2*h)

        rho = 30. * h * delta / np.abs(x1 - x2)
        if rho < 1.:
            h *= rho**(1/4.)
        else:
            if (t + 2.*h) < tmax:
                xs.append(x1)
                ts.append(t + 2*h)
                t += 2*h
            else:
                xs.append(ode_rk4_step(f, xs[i], ts[i], h))
                ts.append(t + h)
                t += h
            i += 1
            h = min(2.*h, h * rho**(1/4.))

    return ts,xs
```

$$\frac{dx}{dt} = -x^3 + \sin t, \qquad x(t=0) = 0.$$




Step size tends to decrease when dx/dt (the r.h.s) is large

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# Stability, stiff equations, and implicit methods

Consider the following ODE

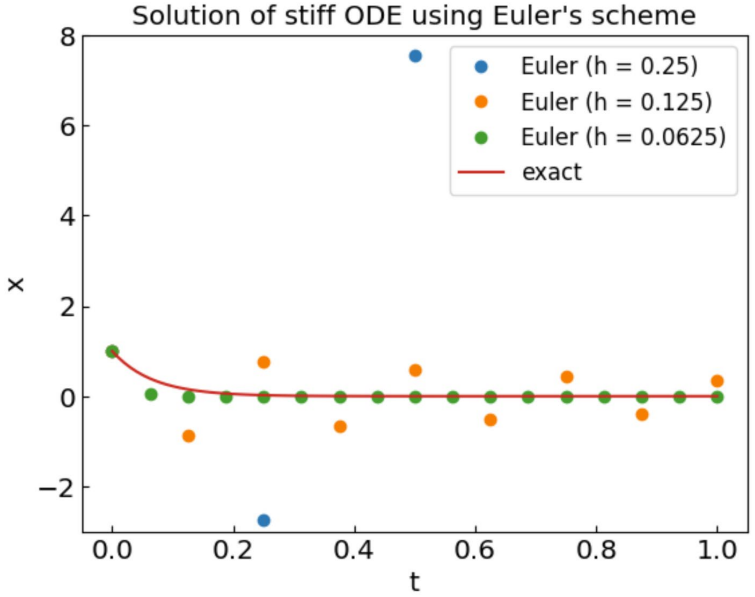
$$\frac{dx}{dt} = -15x, \qquad \textit{stiff equation}$$

with the initial condition *x(t=0)=1*.

The exact solution is of course $x(t) = e^{-15t}$ and goes to zero at large times.

Let us apply Euler's method with h=1/4, 1/8, 1/16

Divergence for h=1/4!

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# Stability, stiff equations, and implicit methods

Consider the following ODE

$$\frac{dx}{dt} = -15x, \qquad \textit{stiff equation}$$

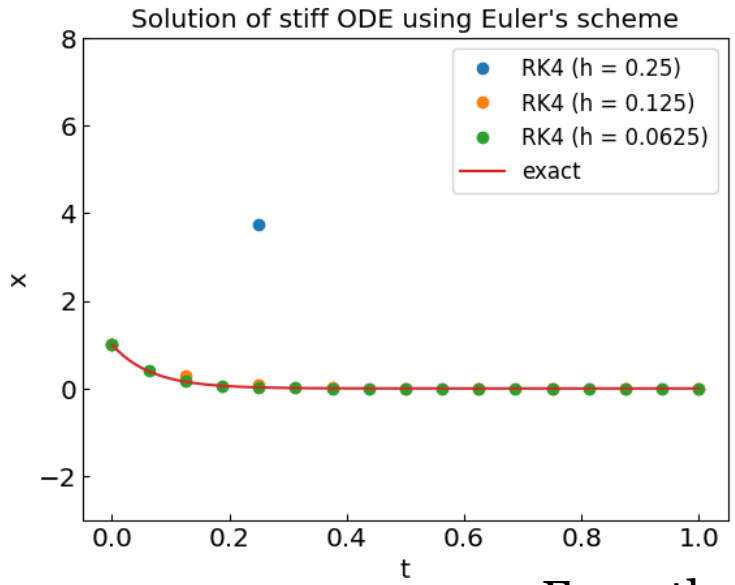
with x(t=0)=1.

The exact solution is of course $x(t) = e^{-15t}$ and goes to zero at large times.

Let us apply Euler's method with h=1/4, 1/8, 1/16

Divergence for h=1/4!

RK4: better but still diverges for h=1/4

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# Euler methods and stiff equations

Recall that in Euler's method x(t+h) = x(t) + h f(x,t)

For $\frac{dx}{dt} = -15x,$ we have $x_{n+1} = x_n - 15hx_n = (1-15h)x_n = (1-15h)^n x_0, \qquad x_n \equiv x(t+nh)$

If |1-15h|>1, i.e. *h*>2/15, the Euler method diverges!

Solution: *implicit methods*

**Implicit Euler method:** $\mathbf{x}(\mathbf{t}+\mathbf{h}) = x(t) + hf[\mathbf{x}(\mathbf{t}+\mathbf{h}), t+h]$

Our stiff equation: $x_{n+1} = x_n - 15hx_{n+1}$ thus $x_{n+1} = \frac{x_n}{1+15h} = \frac{x_0}{(1+15h)^n} \overset{n\to\infty}{\to} 0$ for all $h > 0$.

- Implicit methods are *more stable* than explicit methods
- But require solving non-linear equation for *x(t+h)* at each step
- *Semi-implicit methods:* use one iteration of Newton's method to solve for *x(t+h)*

Other implicit methods: trapezoidal rule, family of implicit Runge-Kutta methods

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# Systems of Ordinary Differential Equations

System of N first-order ODE

$$\frac{dx_1}{dt} = f_1(x_1, \dots, x_N, t),$$
$$\frac{dx_2}{dt} = f_2(x_1, \dots, x_N, t),$$
$$\dots$$
$$\frac{dx_N}{dt} = f_N(x_1, \dots, x_N, t).$$

Vector notation:

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t).$$

- all the methods we covered have the same structure when applied for systems of ODEs
- apply component by component

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# Systems of Ordinary Differential Equations

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t).$$

- Euler method

$$\mathbf{x}(t+h) = \mathbf{x}(t) + h\,\mathbf{f}[\mathbf{x}(t), t]\ .$$

- RK2

$$\begin{aligned}
\mathbf{k_1} &= h\,\mathbf{f}(\mathbf{x}, t), \\
\mathbf{k_2} &= h\,\mathbf{f}(\mathbf{x} + \mathbf{k_1}/2, t + h/2), \\
\mathbf{x}(t+h) &= \mathbf{x}(t) + \mathbf{k_2}\ .
\end{aligned}$$

- RK4

$$\begin{aligned}
\mathbf{k_1} &= h\,\mathbf{f}(\mathbf{x}, t), \\
\mathbf{k_2} &= h\,\mathbf{f}(\mathbf{x} + \mathbf{k_1}/2, t + h/2), \\
\mathbf{k_3} &= h\,\mathbf{f}(\mathbf{x} + \mathbf{k_2}/2, t + h/2), \\
\mathbf{k_4} &= h\,\mathbf{f}(\mathbf{x} + \mathbf{k_3}, t + h), \\
\mathbf{x}(t+h) &= \mathbf{x}(t) + \frac{1}{6}(\mathbf{k_1} + 2\mathbf{k_2} + 2\mathbf{k_3} + \mathbf{k_4})\ .
\end{aligned}$$

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# Systems of Ordinary Differential Equations

```python
def ode_euler_multi(f, x0, t0, h, nsteps):
    """Multi-dimensional version of the Euler method.
    """

    t = np.zeros(nsteps + 1)
    x = np.zeros((len(t), len(x0)))
    t[0] = t0
    x[0,:] = x0
    for i in range(0, nsteps):
        t[i + 1] = t[i] + h
        x[i + 1,:] = ode_euler_step(f, x[i], t[i], h)
    return t,x
```

```python
def ode_rk2_multi(f, x0, t0, h, nsteps):
    """Multi-dimensional version of the RK2 method.
    """

    t = np.zeros(nsteps + 1)
    x = np.zeros((len(t), len(x0)))
    t[0] = t0
    x[0,:] = x0
    for i in range(0, nsteps):
        t[i + 1] = t[i] + h
        x[i + 1] = ode_rk2_step(f, x[i], t[i], h)
    return t,x
```

```python
def ode_rk4_multi(f, x0, t0, h, nsteps):
    """Multi-dimensional version of the RK4 method.
    """

    t = np.zeros(nsteps + 1)
    x = np.zeros((len(t), len(x0)))
    t[0] = t0
    x[0,:] = x0
    for i in range(0, nsteps):
        t[i + 1] = t[i] + h
        x[i + 1] = ode_rk4_step(f, x[i], t[i], h)
    return t,x
```

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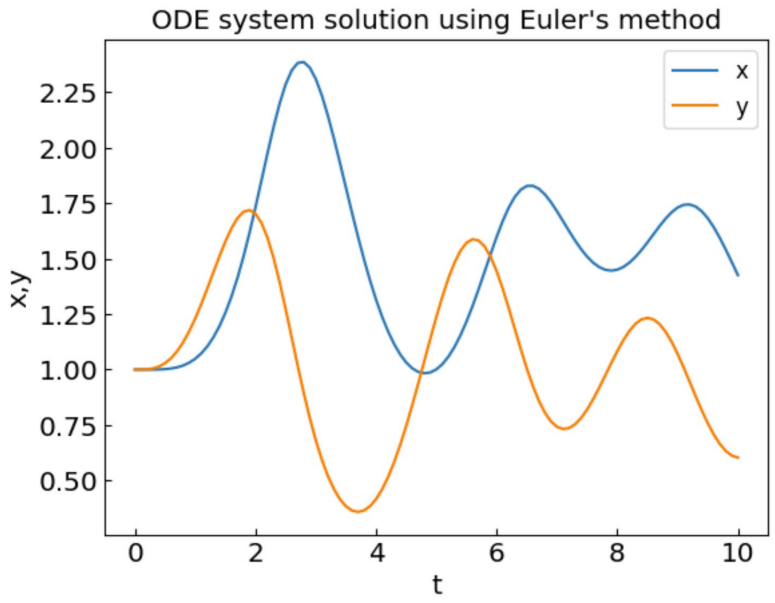
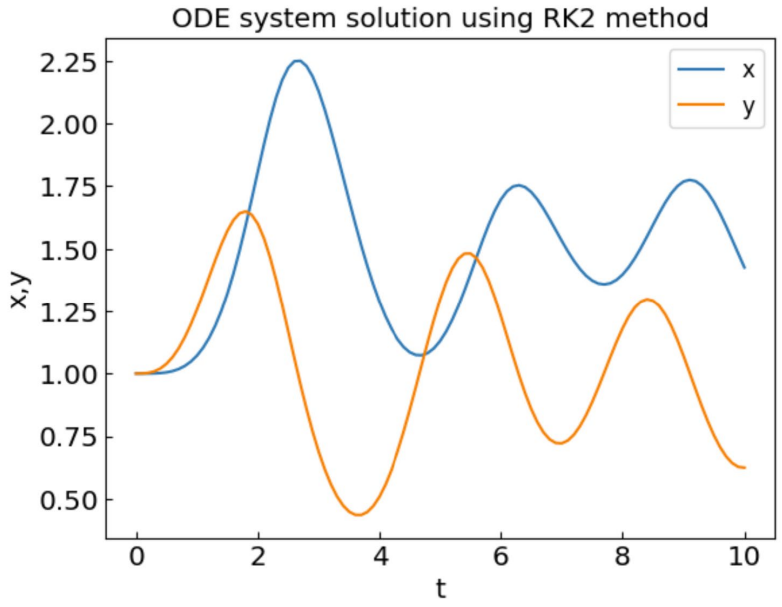
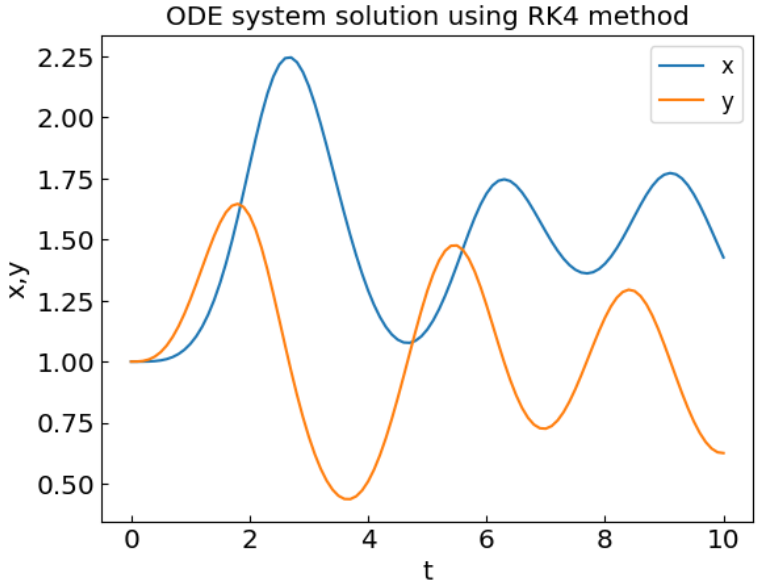
# Systems of Ordinary Differential Equations: Example

$$\frac{dx}{dt} = xy - x,$$
$$\frac{dy}{dt} = y - xy + (\sin t)^2$$

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# Systems of 2nd-order ODEs

Newton/Lagrange equations of motion are 2nd order systems of ODE

$$m_i \frac{d^2 x_i}{dt^2} = F_i(\{x_j\}, \{dx_j/dt\}, t)$$

A system of *N* second-order ODEs

$$\frac{d^2\mathbf{x}}{dt^2} = \mathbf{f}(\mathbf{x}, d\mathbf{x}/dt, t),$$

can be written as a system of *2N* first-order ODEs by denoting $\frac{d\mathbf{x}}{dt} = \mathbf{v}$

$$\frac{d\mathbf{x}}{dt} = \mathbf{v},$$
$$\frac{d\mathbf{v}}{dt} = \mathbf{f}(\mathbf{x}, \mathbf{v}, t),$$

and can be solved for **x**(t) and **v**(t) using standard methods

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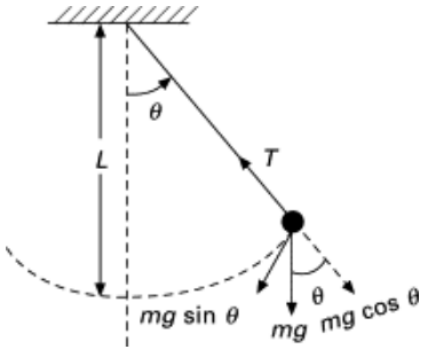
# Example: Simple pendulum

The equation of motion for a simple pendulum reads

$$mL\frac{d^2\theta}{dt^2} = -mg\sin\theta.$$




denote $\frac{d\theta}{dt} = \omega$ and write a system of two first-order ODE

$$\frac{d\theta}{dt} = \omega,$$
$$\frac{d\omega}{dt} = -\frac{g}{L}\sin\theta\,,$$

For small angles $\sin\theta \approx \theta$ , an analytic solution exists

$$\theta(t) \approx \theta_0 \cos\left(\sqrt{\frac{g}{L}}t + \phi\right)$$

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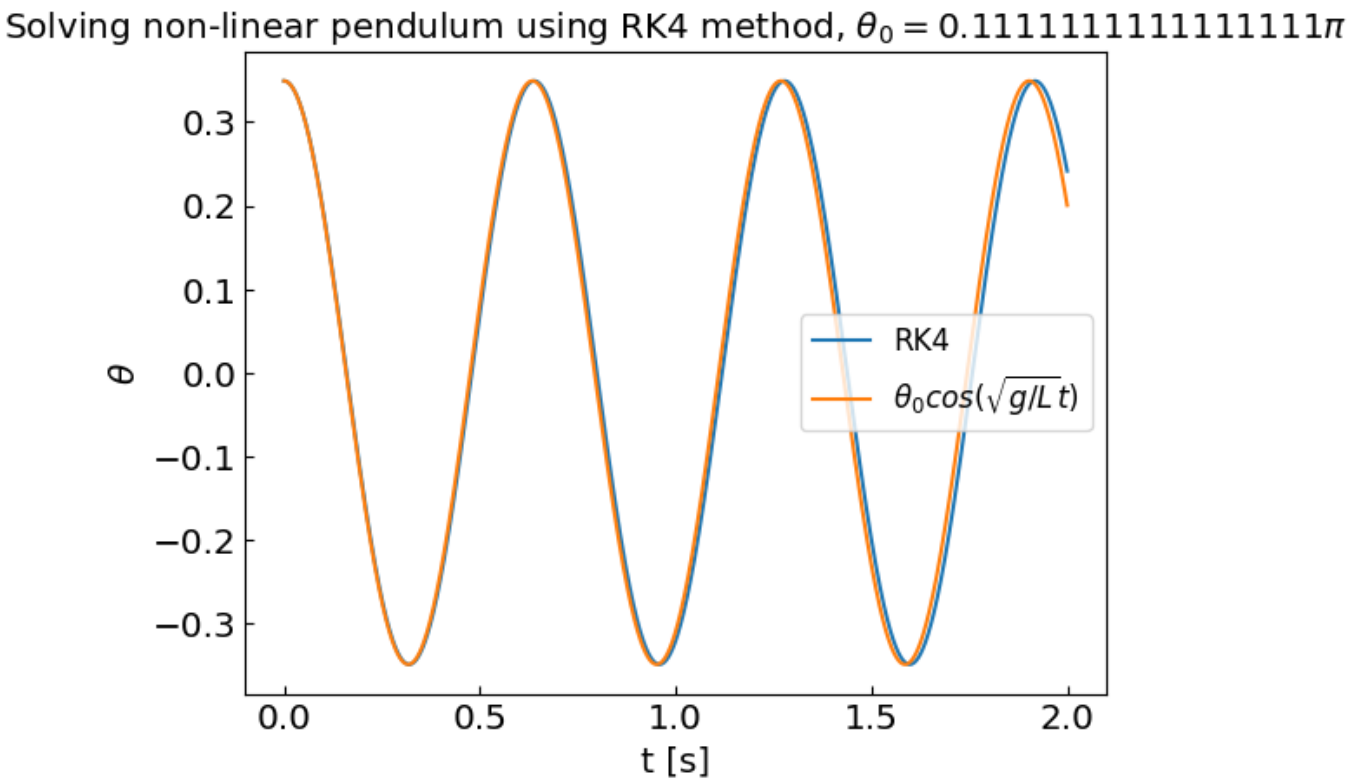
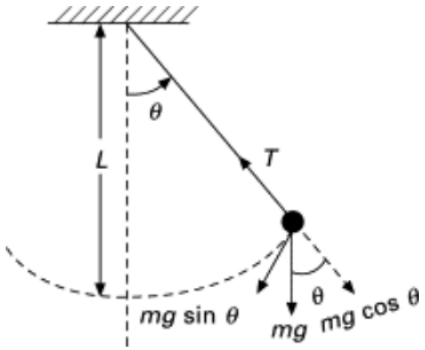
# Example: Simple pendulum

Initially at rest at angle $\theta_0 = 20° \approx 0.111\pi$ L=0.1 m, g=9.81 m/s$^2$




Linear regime at small angles

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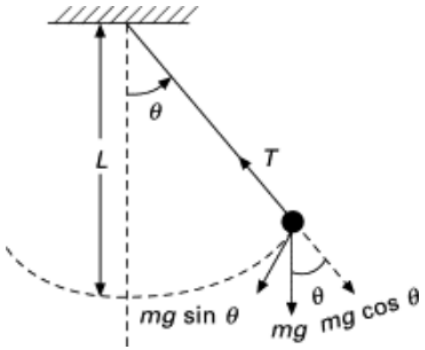
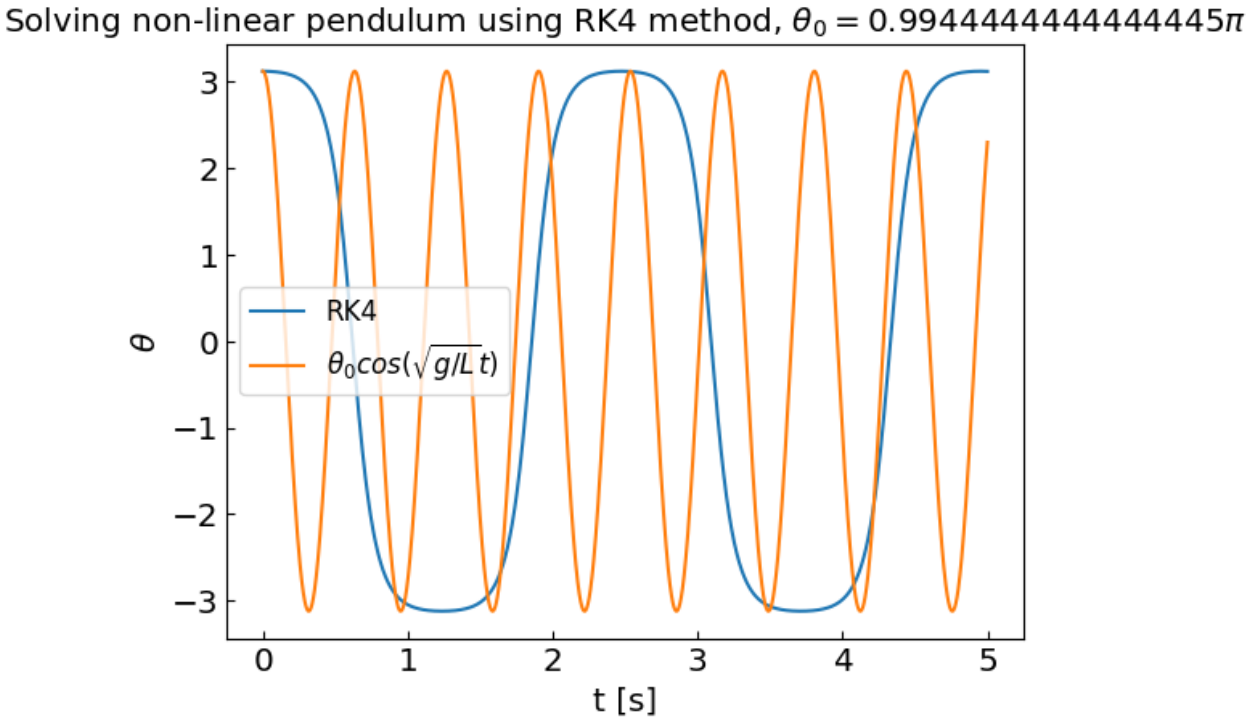
# Example: Simple pendulum

Initially at rest at angle $\theta_0 = 179° \approx 0.994\pi$ L=0.1 m, g=9.81 m/s$^2$




Non-linear regime at large angles, approximate analytic solution fails

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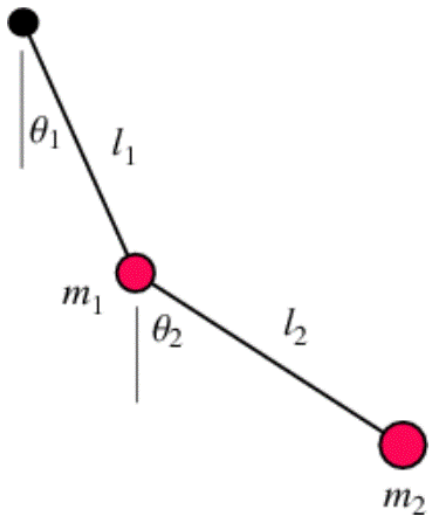
# Double pendulum

Double pendulum is the simplest system exhibiting chaotic motion – *deterministic chaos*




Two degrees of freedom: the displacement angles $\theta_1$ and $\theta_2$.

$$
\begin{aligned}
x_1 &= l_1 \sin(\theta_1), \\
y_1 &= -l_1 \cos(\theta_1), \\
x_2 &= l_1 \sin(\theta_1) + l_2 \sin(\theta_2), \\
y_2 &= -l_1 \cos(\theta_1) - l_2 \cos(\theta_2).
\end{aligned}
$$

$$T = \frac{m_1 \dot{x}_1^2}{2} + \frac{m_2 \dot{x}_2^2}{2} = \frac{1}{2} m_1 l_1^2 \dot{\theta}_1 + \frac{1}{2} m_2 \left[ l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2), \right]$$ kinetic energy

$$V = m_1 g y_1 + m_2 g y_2 = -(m_1 + m_2) g l_1 \cos(\theta_1) - m_2 g l_2 \cos(\theta_2).$$ potential energy

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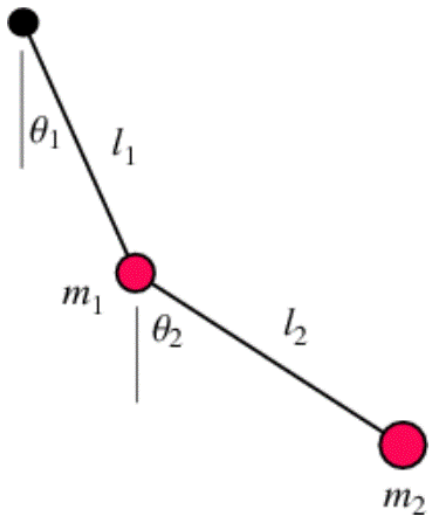


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The Lagrange equations of motion read

$$
\begin{aligned}
(m_1 + m_2) l_1 \ddot{\theta}_1 + m_2 l_2 \cos(\theta_1 - \theta_2) \ddot{\theta}_2 &= -m_2 l_1 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) - (m_1 + m_2) g \sin(\theta_1), \\
m_2 l_1 \cos(\theta_1 - \theta_2) \ddot{\theta}_1 + m_2 l_2 \ddot{\theta}_2 &= m_2 l_1 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) - m_2 g \sin(\theta_2).
\end{aligned}
$$

This is a system of two linear equation for $\ddot{\theta}_{1,2}$ that can be solved easily.

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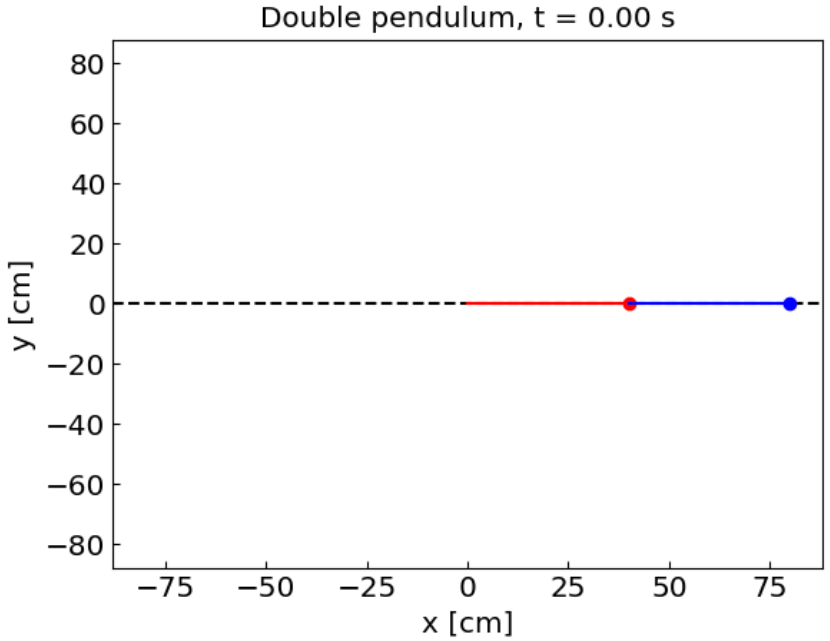
# Double pendulum

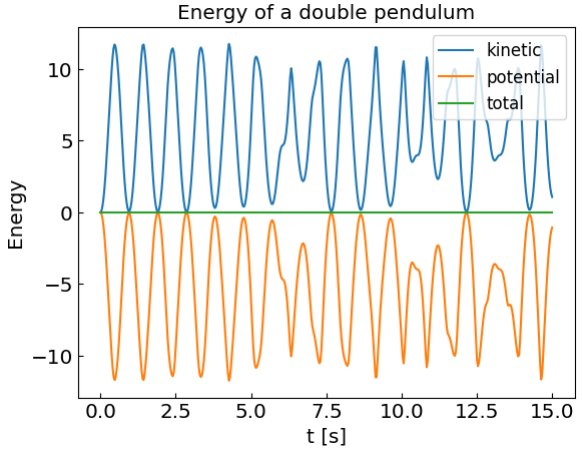
```
g = 9.81
l1 = 0.4
l2 = 0.4
m1 = 1.0
m2 = 1.0

def fdoublependulum(xin, t):
    global f_evaluations
    f_evaluations += 1

    theta1 = xin[0]
    theta2 = xin[1]
    omega1 = xin[2]
    omega2 = xin[3]
    a1 = (m1 + m2)*l1
    b1 = m2*l2*np.cos(theta1 - theta2)
    c1 = m2*l2*omega2*omega2*np.sin(theta1 - theta2) + (m1 + m2)*g*np.sin(theta1)
    a2 = m2*l1*np.cos(theta1 - theta2)
    b2 = m2*l2;
    c2 = -m2*l1*omega1*omega1*np.sin(theta1 - theta2) + m2*g*np.sin(theta2) # + k
    domega1 = - ( c2/b2 - c1/b1 ) / ( a2/b2 - a1/b1 )
    domega2 = - ( c2/a2 - c1/a1 ) / ( b2/a2 - b1/a1 )
    return np.array([omega1,
                     omega2,
                     domega1,
                     domega2
                    ])
```

```
sol = bulirsch_stoer(fpendulum, x0, t0, 1, tend, eps, error_definition_pendulum, 10)
```

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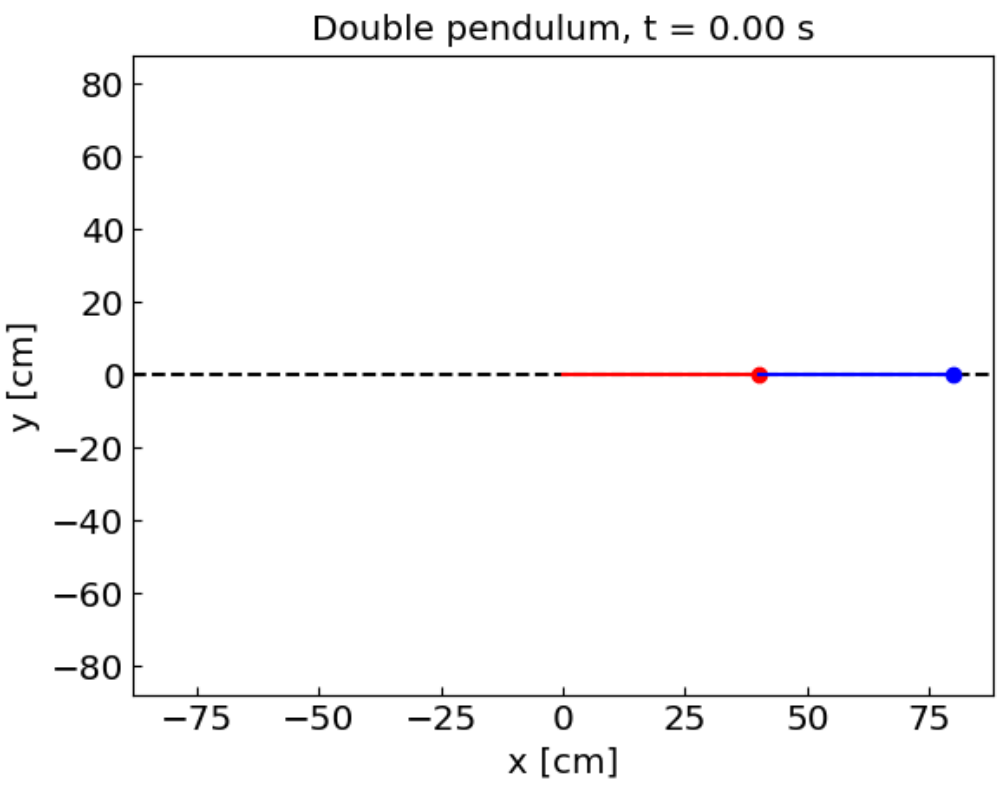
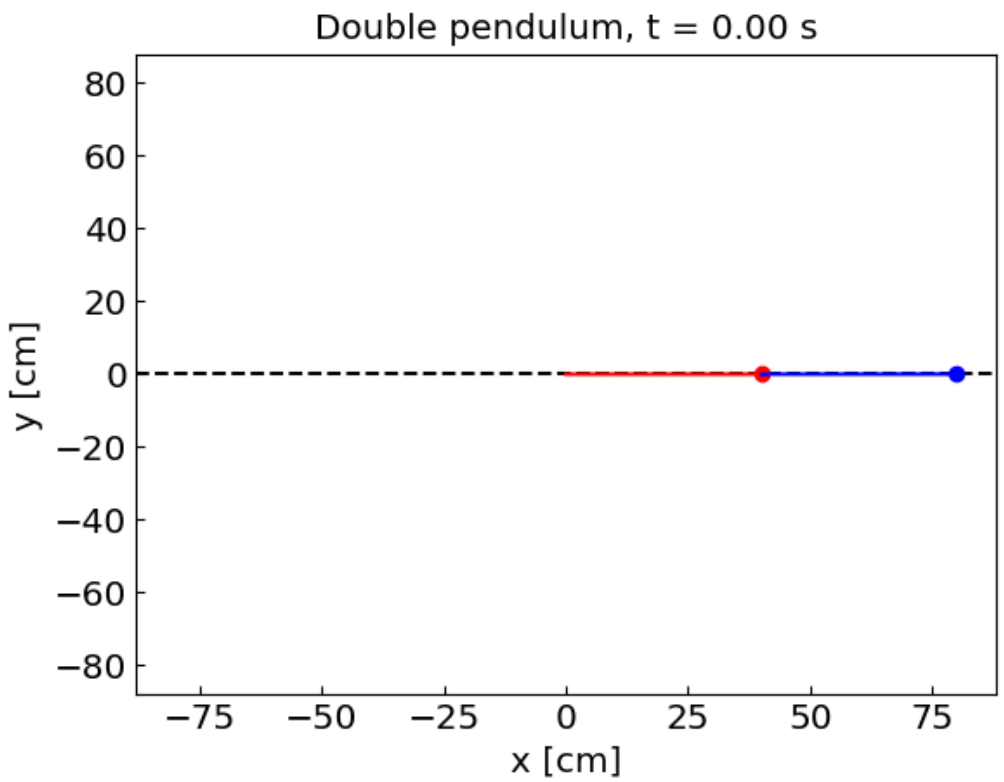
# Double pendulum: chaotic behavior

$$\theta_1^0 = \theta_2^0 = \pi/2$$




$$\theta_1^0 = \theta_2^0 = \pi/2 + 10^{-4}$$

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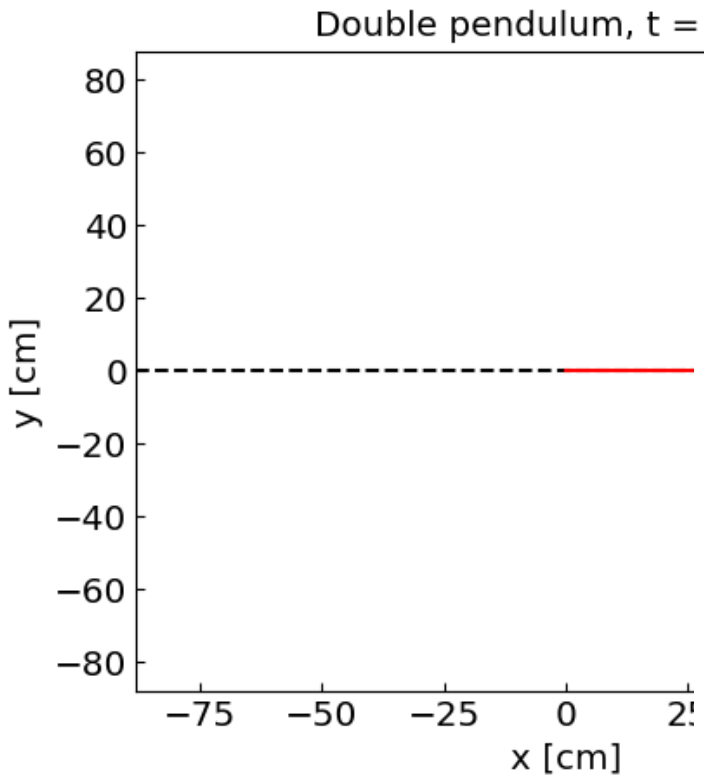


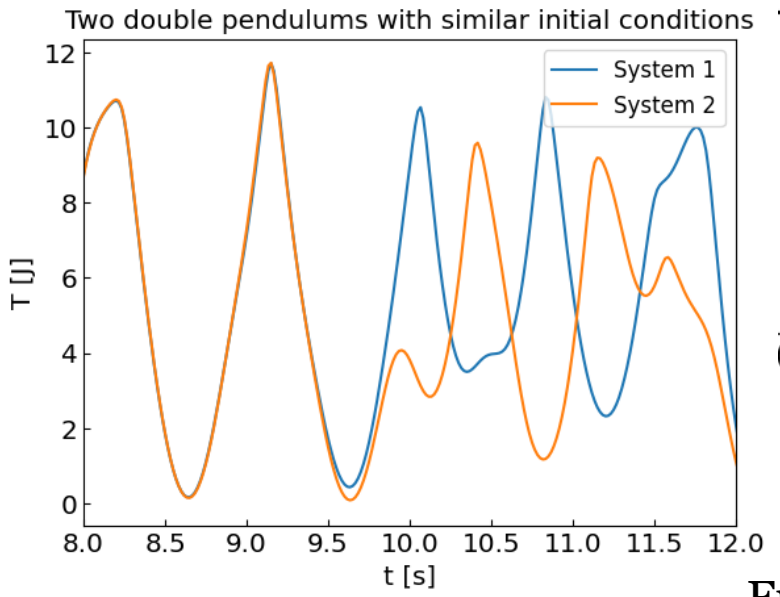


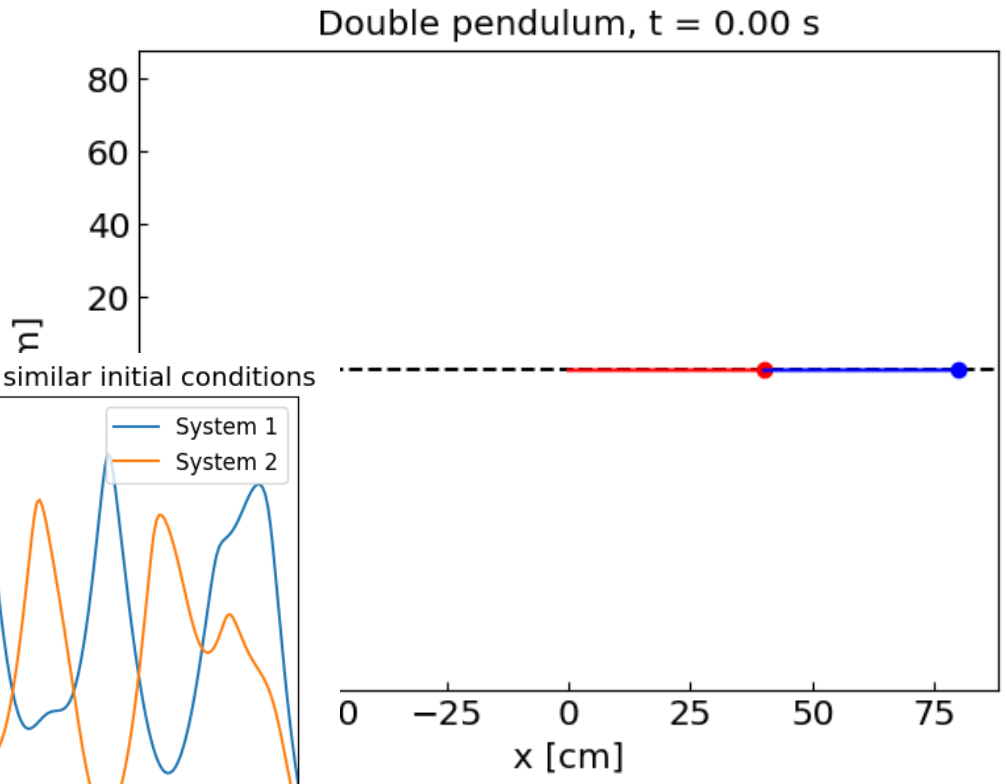


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# Three-body problem

Three stars interacting through gravitational force

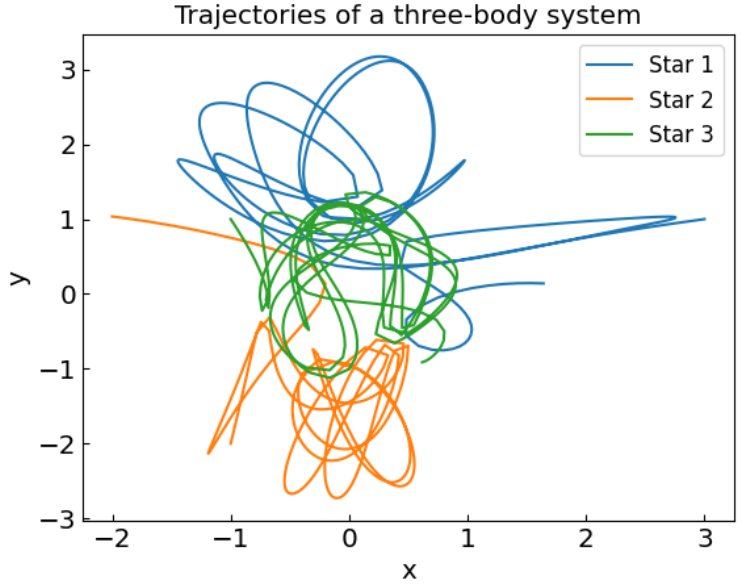
The equations of motion are

$$\frac{d^2\mathbf{r}_1}{dt} = Gm_2\frac{\mathbf{r}_2-\mathbf{r}_1}{|\mathbf{r}_2-\mathbf{r}_1|^3} + Gm_3\frac{\mathbf{r}_3-\mathbf{r}_1}{|\mathbf{r}_3-\mathbf{r}_1|^3},$$

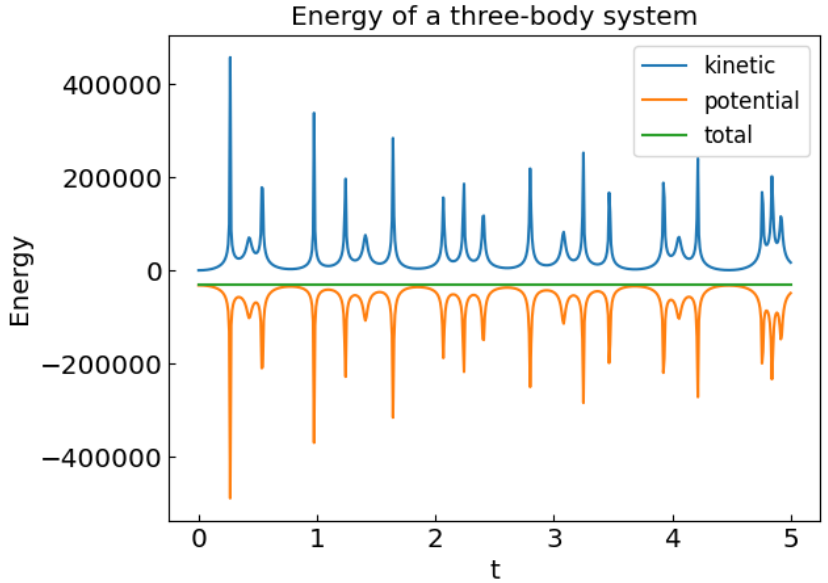
$$\frac{d^2\mathbf{r}_2}{dt} = Gm_1\frac{\mathbf{r}_1-\mathbf{r}_2}{|\mathbf{r}_1-\mathbf{r}_2|^3} + Gm_3\frac{\mathbf{r}_3-\mathbf{r}_2}{|\mathbf{r}_3-\mathbf{r}_2|^3},$$

$$\frac{d^2\mathbf{r}_3}{dt} = Gm_1\frac{\mathbf{r}_1-\mathbf{r}_3}{|\mathbf{r}_1-\mathbf{r}_3|^3} + Gm_2\frac{\mathbf{r}_2-\mathbf{r}_3}{|\mathbf{r}_2-\mathbf{r}_3|^3}.$$

| Name | Mass | x | y |
|---|---|---|---|
| Star 1 | 150. | 3 | 1 |
| Star 2 | 200. | -1 | -2 |
| Star 3 | 250. | -1 | 1 |




Cannot be solved analytically!




Take $G = 1$ (dimensionless)

Initially at rest and move in plane z = 0

*Initial coordinates:* $\mathbf{r}_1 = (3,1)$, $\mathbf{r}_2 = (-1,-2)$, $\mathbf{r}_3 = (-1,1)$

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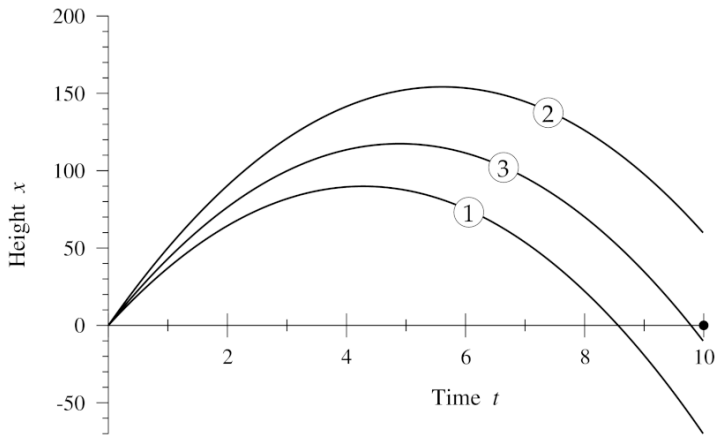
# Boundary value problems and the shooting method

Sometimes we have equations, such as vertically thrown object

$$\frac{dx}{dt} = v,$$
$$\frac{dv}{dt} = -g,$$

and boundary conditions, e.g. x(0) = 0 and x(10) = 0 instead of initial conditions $v(0) = v_0$.




How to solve this problem?

In the shooting method one takes trial values of $v_0$ until finding the one where the solution satisfies the boundary condition x(10) = 0.

To find $v_0$ efficiently one combines numerical ODE method (e.g. RK4) with non-linear equation solver (e.g. bisection method).

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